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Compositions and methods for making benzyloquinoline alkaloids, morphinan alkaloids, thebaine, and derivatives thereof

Abstract

Disclosed herein are methods that may be used for the synthesis of benzyloquinoline alkaloids (“BIAs”) such as alkaloid morphinan. The methods disclosed can be used to produce thebaine, oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, and/or buprenorphine. Compositions and organisms useful for the synthesis of BIAs, including thebaine synthesis polypeptides, purine permeases, and polynucleotides encoding the same, are provided.

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Background/Summary

CROSS-REFERENCE (1) This application is a Continuation of U.S. application Ser. No. 16/312,776, filed on Dec. 21, 2018, U.S. patent Ser. No. 11/142,780, which is a 371 of International Application No. PCT/US2017/039589, filed on Jun. 27, 2017 which claims the priority benefit of U.S. Provisional Patent Application Ser. Nos. 62/355,022, filed Jun. 27, 2016; 62/433,431, filed Dec. 13, 2016; 62/438,540, filed Dec. 23, 2016; 62/438,601, filed Dec. 23, 2016; 62/438,702, filed Dec. 23, 2016; 62/438,588, filed Dec. 23, 2016; 62/438,695, filed Dec. 23, 2016; 62/469,006, filed Mar. 9, 2017; and 62/514,104, filed on Jun. 2, 2017, all of which are hereby incorporated herein by reference in their entirety.

SEQUENCE LISTING

(1) The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. The ASCII copy was created on Mar. 16, 2022, Sep. 24, 2019, is named 125US12 SequenceListing.txt, and is 114,679 bytes in size.

BACKGROUND OF THE DISCLOSURE

(2) The following paragraphs are provided by way of background and not an admission that anything discussed therein is prior art or part of the knowledge of persons skilled in the art.

(3) Thebaine, a chemical compound also known as paramorphine and codeine methyl enol ether, belongs to the class of compounds known as benzyloisoquinoline alkaloids ("BIAs"), and within that class, to a BIA subclass of compounds known as morphinan alkaloids, and has long been recognized as a useful feedstock compound in the manufacture of therapeutic agents, including, for example, morphine and codeine. Other agents that can be manufactured can include but are not limited to oripavine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone and neopinone. It is known that thebaine in planta is produced from a precursor compound named salutaridine via intermediate morphinan alkaloids named salutaridinol and salutaridinol 7-O-acetate. Currently thebaine may be harvested from natural sources, such as opium poppy capsules (see e.g., U.S. Pat. Appl. Pub. No 2002/0106761; see also e.g., *Poppy*, the genus *Papaver*, 1998, pp 113, Harwood Academic Publishers, Editor: Bernáth, J.). Alternatively,

thebaine may be prepared synthetically. The latter may be achieved by a reaction sequence starting with ketalization of iodoisovanillin (see e.g., Rinner, U. and Hudlicky, T., 2012, Top. Cur. Chem. 309: 33-66; Stork, G., 2009, J. Am. Chem. Soc. 131 (32) pp 11402-11406).

(4) The existing manufacturing methods for BIAs, including thebaine and other morphinan alkaloids and their derivatives, suffer from low yields and/or are expensive. Some of the known methodologies for the manufacture of thebaine exist in the production of undesirable quantities of morphinan alkaloid by-products (see e.g., Rinner, U., and Hudlicky, J., 2012, Top. Cur. Chem. 209: 33-66). No methods exist to commercially biosynthetically manufacture BIAs, including thebaine and other morphinan alkaloids and their derivatives. Therefore, there is a need for improved methods for the synthesis of BIAs including thebaine and other morphinan alkaloids and their derivatives.

INCORPORATION BY REFERENCE

(5) All publications, patents, and patent applications herein are incorporated by reference to the same extent as if each individual publication, patent, or patent application was specifically and individually indicated to be incorporated by reference. In the event of a conflict between a term herein and a term in an incorporated reference, the term herein controls.

SUMMARY

(6) This application discloses certain alkaloids belonging to the class of benzyloisoquinoline alkaloids ("BIAs"), morphinan alkaloids, and their derivatives, as well as to methods of making such BIAs, including morphinan alkaloids (and their derivatives). For example, one BIA that can be made is thebaine.

(7) BIAs, such as morphinan alkaloids, more specifically thebaine, can be produced from sugar (or other substrates). Disclosed herein is a cell (e.g., a microorganism) that comprises one or more nucleic acids encoding for heterologous enzymes that can perform any one of the following reactions: i) sugar to 1-tyrosine; ii) 1-tyrosine to 1-DOPA; iii) 1-DOPA to dopamine; iv) dopamine to (S)-norcoclaurine; v) (S)-norcoclaurine to (S)/(R)-reticuline; vi) (R)-reticuline to salutaridine; vii) salutaridine to salutaridinol; viii) salutaridinol to salutaridinol-7-O-acetate; ix) salutaridinol-7-O-acetate to thebaine; x) thebaine to oripavine; morphinone; morphine; codeine; codeinone; and/or neopinone. In some instances, the enzyme that converts salutaridinol-7-O-acetate to thebaine is a thebaine synthesis polypeptide (a.k.a. thebaine synthase or BetV1). In some instances, a purine permease is also present within the cell.

(8) The method of converting a sugar (or other substrate) into a BIA, such as thebaine, can also be performed in vitro. The one or more enzymes that can perform any one of the following reactions: i) sugar to 1-tyrosine; ii) 1-tyrosine to 1-DOPA; iii) 1-DOPA to dopamine; iv) dopamine to (S)-norcoclaurine; v) (S)-norcoclaurine to (S)/(R)-reticuline; vi) (R)-reticuline to salutaridine; vii) salutaridine to salutaridinol; viii) salutaridinol to salutaridinol-7-O-acetate; ix) salutaridinol-7-O-acetate to thebaine; x) thebaine to oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, and/or buprenorphine; can be isolated (or contained in a cellular extract) and used in an in vitro reaction. In some cases, some of the reactions can be performed in vivo (e.g., within a cell), while others can be performed in vitro. For example, sugar can be converted to salutaridine in vivo, while the reaction of salutaridine to thebaine can occur in vitro. Should thebaine be the end product, a thebaine synthesis polypeptide can be used to convert salutaridinol-7-O-acetate to thebaine in vitro.

(9) In cases where a thebaine synthesis polypeptide is used, a *Papaver* thebaine synthesis polypeptide can be chosen. For example, the thebaine synthesis polypeptide can be a *Papaver somniferum* thebaine synthesis polypeptide. The thebaine synthesis polypeptide can comprise an amino acid sequence substantially identical to SEQ ID NO. 6. In some cases, the thebaine synthesis polypeptide can comprise an amino acid sequence substantially identical SEQ ID NO. 31.

(10) In cases where a purine permease is used, a purine permease from the genus *Papaver* can be chosen. For example, the purine permease can be from a *Papaver somniferum*. The purine

permease can comprise a nucleotide sequence substantially identical to SEQ ID NO. 36. The purine permease can comprise a nucleotide sequence substantially identical to SEQ ID NO. 38. The purine permease can comprise a nucleotide sequence substantially identical to any one of SEQ ID NOs. 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, or 64. In some cases, the purine permease can comprise an amino acid sequence substantially identical SEQ ID NO. 35. In some cases, the purine permease can comprise an amino acid sequence substantially identical SEQ ID NO. 37. In some cases, the purine permease can comprise an amino acid sequence substantially identical to any one of SEQ ID NOs. 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, or 63.

(11) In some cases, the cell disclosed herein can be a prokaryote. The cell can also be a eukaryote in some cases. Other microorganisms that can be used are *Saccharomyces cerevisiae*, *Yarrowia lipolytica*, or *Escherichia coli*.

(12) Also disclosed herein are vectors or chimeric polynucleotides comprising one or more polynucleotides capable of controlling expression in a host cell which is operably linked to a polynucleotide encoding a thebaine synthesis polypeptide. In some cases, a polynucleotide encoding for a purine permease can be within the vector or chimeric polynucleotides.

(13) Another method disclosed herein is a screening method for modulating expression of polynucleotide sequences in a cell expressing thebaine synthesis polypeptide (and/or a purine permease). The screening method can comprise (a) mutagenizing a cell expressing thebaine synthesis polypeptide (and/or purine permease); (b) growing the cell to obtain a plurality of cells; and determining if a cell from has modulated levels of thebaine synthesis polypeptide (and/or purine permease) or if the thebaine synthesis polypeptide (and/or purine permease) has increased/decreased levels of activity compared to a non-mutagenized cell.

(14) Other substrates besides sugar can be used to make BIAs. For example, the method or the microorganism can use sugar, glycerol, L-tyrosine, tyramine, L-3,4-dihydroxyphenyl alanine (L-DOPA), an alcohol, dopamine, or combination thereof, as a substrate to make BIAs.

(15) The BIAs such as thebaine and other morphinan alkaloids and their derivatives obtained from the cells and/or methods disclosed can be used as a feedstock compound in the manufacture of a medicinal compound.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The disclosure is in the hereinafter provided paragraphs described in relation to its Figures. The Figures provided herein are provided for illustration purposes and are not intended in any way to be limiting.

(2) FIGS. 1A-1D. FIG. 1A depicts the pathway from glucose to L-tyrosine in yeast. TKT, transketolase; E4P, erythrose 4-phosphate; (PEP) phosphoenolpyruvic acid; PEPS (PEP synthetase; DAHPS, DAHP synthase; DAHP, 3-deoxy-D-arabinoheptulosonate 7-phosphate; CM/PDH, chorismate mutase/prephenate dehydrogenase; HPP, hydroxyphenylpyruvate. FIG. 1B depicts the pathway from glucose to L-tyrosine in *E. coli*. XSP, xylulose 5-phosphate; PYR, pyruvate; EPSP, 5-enolpyruvoylshikimate 3-phosphate; CHA, chorismate; PPA, prephenate; HPP, 4-hydroxyphenylpyruvate; L-Glu, glutamic acid; and α -KG, α -ketoglutarate. The enzymes (in boldface) are as follows: PpsA, phosphoenolpyruvate synthase; TktA, transketolase A; AroG, DAHP synthase; AroB, DHQ synthase; AroD, DHQ dehydratase; YdiB, quinate/shikimate dehydrogenase; AroE, shikimate dehydrogenase; AroK/L, shikimate kinase I/II; AroA, EPSP synthase; AroC, chorismate synthase; TyrA, chorismate mutase/prephenate dehydrogenase; and TyrB, tyrosine aminotransferase. QUIN and gallic acid (GA) are side products. QUIN is formed by YdiB from DHQ, while GA is formed by AroE from DHS. The dashed lines indicate where feedback inhibitions occur. FIG. 1C. depicts the pathway from L-tyrosine to thebaine and other

morphinan alkaloids such as codeine. FIG. 1D shows another depiction of the biosynthetic pathway from tyrosine to morphine in opium poppy. The enzymes listed are as follows: TYDC, tyrosine/DOPA decarboxylase; TyrAT, tyrosine aminotransferase; NCS, norcoclaurine synthase; 6OMT, norcoclaurine 6-O-methyltransferase; CNMT, coclaurine N-methyltransferase; NMCH, N-methylcoclaurine 3'-hydroxylase; 3'-hydroxyl-N-methylcoclaurine 4'-O-methyltransferase; REPI, reticuline epimerase; SalSyn, salutaridine synthase; SalR, salutaridine reductase; SalAT, salutaridinol 7-O-acetyltransferase; THS, thebaine synthesis polypeptide (a.k.a., thebaine synthase or Betv1); T6ODM, thebaine 6-O-methyltransferase, COR, codeinone reductase; CODM, codeine O-demethylase.

(3) FIG. 2 depicts a prototype chemical structure of a morphinan alkaloid. Various atoms within the structures have been numbered for ease of reference.

(4) FIGS. 3A-C depicts the chemical structures of example substrate morphinan alkaloids, notably salutaridine (FIG. 3A), salutaridinol (FIG. 3B), and salutaridinol 7-O-acetate (FIG. 3C).

(5) FIG. 4 depicts the chemical structure of the morphinan alkaloid compound thebaine.

(6) FIG. 5 depicts a chemical reaction illustrating the formation of thebaine in the presence of thebaine synthesis polypeptide (THS), and three morphinan alkaloid non-enzymatic by-products, referred to, as denoted, as Product #1, Product #2 and (7R)-salutaridinol, in the absence of THS. Compounds with m/z 330 are alternative reaction products of (7S)-salutaridinol-7-O-acetate, and are not thebaine.

(7) FIGS. 6A-6G. FIG. 6A shows that thebaine formation is enhanced using latex protein extracts in SalAT-coupled assays. Using SalAT alone, a major product with m/z 330 and little thebaine were formed. With the addition of latex protein extracts, thebaine (m/z 312) formation increased substantially and the relative abundance of the m/z 330 compound decreased. The additional mannitol during latex protein extraction further enhanced thebaine formation. Denatured latex protein was as a negative control. FIG. 6B shows the LC-MS.sub.n characterization of the m/z 330 by-product(s) of SalAT-coupled assays performed in the absence of latex protein extracts. FIG. 6C depicts the proposed identity and reaction mechanism leading to the formation of the m/z 330 by-product(s) of SalAT-coupled assays performed in the absence of latex protein extracts. Either or all of the putative m/z 330 by-products could be formed. FIGS. 6D-6F show the elution profiles for hydrophobic interaction chromatography (HIC) on a butyl-S sepharose column (FIG. 6D), ion-exchange (IEX) on a HiTrap Q column (FIG. 6E), and size-exclusion chromatography (SEC) on a Superdex 75 HR 10/30 column (FIG. 6F). Elution fractions containing thebaine-forming protein active in SalAT-coupled assays are shown as grey bars. Fractions 10-13 from the SEC step were pooled and subjected to proteomics analysis. FIG. 6G shows a partial purification table for the thebaine-forming protein active in SalAT-coupled assays.

(8) FIG. 7A-7G. FIG. 7A depicts certain experimental results, notably relative amounts of thebaine produced by SalAT in the presence of several protein fractions (AS, HIC, IEC and HA fractions) from the purification process, as further described in Example 1. Also shown is a polyacrylamide gel following gel electrophoresis of the same protein extracts. FIGS. 7B-7D depict an SDS-PAGE for of each step in the purification of the thebaine-forming protein active in SalAT-coupled assays. FIG. 7B show crude latex protein, ammonium sulphate (AS) precipitate, and HIC and IEX chromatographic fractions resolved on 10% SDS-PAGE gels. FIG. 7C show fractions 10-13 from the SEC step resolved on a 10% SDS-PAGE gel. FIG. 7D show the combined SEC fractions 10-13 resolved on a 14% SDS-PAGE gel. FIG. 7E shows the top six candidates for the thebaine-forming protein active in SalAT-coupled assays from proteomics analysis of the three bands resolved on a 14% SDS-PAGE gel of the combined SEC fractions 10-13 (FIG. 7D). FIG. 7F depicts a protein sequence alignment of the top six candidates for the thebaine-forming protein active in SalAT-coupled assays. Betv1-1 (SEQ ID NO. 6) is also referred to as thebaine synthesis polypeptide or thebaine synthase throughout the disclosure. FIG. 7G depicts certain experimental results, notably two polyacrylamide gels following gel electrophoresis of protein extracts (SEC and HA fractions)

and purification of the 16-18 kDa bands after inclusion of the SEC step before the hydroxyapatite (HA) step, with retention of the thebaine synthesis protein (TS) activity, as further described in Example 1.

(9) FIGS. 8A and 8B shows SalAT-coupled enzyme assays of recombinant proteins produced in *Escherichia coli* representing the top six candidates responsible for the enhanced conversion of salutaridinol to thebaine. Betv1-1 is also referred to as thebaine synthesis polypeptide or thebaine synthase throughout the disclosure. FIG. 8A shows His.sub.6-tag affinity-purified candidate proteins resolved by SDS-PAGE. FIG. 8B shows LC-MS chromatographs in SalAT-coupled assays showing enhanced thebaine (m/z 312) and reduced m/z 330 by-product formation by one candidate, Betv1-1.

(10) FIGS. 9A-D. FIGS. 9A-9C depict the testing of the top six candidates for the thebaine-forming protein active in *Saccharomyces cerevisiae* cultures fed salutaridine for 24 hours. FIG. 9A depicts the pathway showing the conversion of salutaridine to thebaine by the successive actions of SalR, SalAT and thebaine synthesis polypeptide (THS). Betv1-1 is also referred to as thebaine synthesis polypeptide or thebaine synthase throughout the disclosure. FIG. 9B depicts an immunoblot showing the occurrence of all six candidate proteins in independent yeast strains.

(11) FIG. 9C shows thebaine formation in engineered yeast strains expressing the six candidate genes. Empty vector refers to a yeast strain expressing only SalR and SalAT. FIG. 9D shows thebaine production titers from extracts of *E. coli* expressing SalAT, Bet v1-1A, Bet v1-1B, PR10-3A, PR10-3B, and all constructs together. Bet v1-1A refers to a C-terminal HA tagged Betv1, whereas Bet v1-1B corresponds to an N-terminal HA tagged Betv1. Bet v1-1B produced the most thebaine from single construct extracts whereas Bet v1-1A produced slightly less. Extracts from *E. coli* expressing all constructs together produced the highest titers of thebaine.

(12) FIG. 10 shows that the majority of salutaridinol was not consumed in extracts of *E. coli* expressing Bet v1-1A, Bet v1-1B, PR10-3A, PR10-3B, and all constructs together. However, with SalAT alone, all salutaridinol was used and formed m/z 330 byproduct. Bet v1-1A refers to a C-terminal HA tagged Betv1, whereas Bet v1-1B corresponds to an N-terminal HA tagged Betv1.

(13) FIGS. 11A-D. FIG. 11A shows the genomic organization of genes encoding two isoforms of reticuline epimerase (REPI), salutaridine synthase (SalSyn), salutaridine reductase (SalR), salutaridine 7-O-acetyltransferase (SalAT), and thebaine synthesis polypeptide (THS) as present in the *Papaver somniferum*. Several other uncharacterized genes are also physically linked to the cluster of genes involved in the conversion of (S)-reticuline to thebaine. THS1 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 80; and THS2 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 6. FIG. 11B shows the alignment of THS1 and THS2 isoforms. THS1 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 80; THS1s is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 81; THS2 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 6; and THS2s is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 32. FIG. 11C shows a proteomics analysis of protein extracts derived from whole stem, latex-free stem, and latex isolated from the opium poppy T chemotype. FIG. 11D depicts a phylogenetic tree of selected PR10-family proteins from plants. Characterized PR10 proteins with resolved 3D structures were selected from *Arabidopsis thaliana*, *Betula pendula*, *Lupinus luteus*, *Thalictrum flavum*, and *Vigna radiata*. Uncharacterized THS orthologs were selected from related *Papaver* species including *Papaver bacterium*, *Papaver rhoeas*, and *Papaver setigerum*. An alignment was constructed using CLUSTALW with a cost matrix of BLOSUM. The phylogenetic tree was then built on the basis of the alignment using PHYML with a substitution model of LG and 100 bootstraps.

(14) FIGS. 12A-12C. FIG. 12A is a schematic representation showing the location of primers used for the detection of specific THS genes and splice-variants. The location of fragments used for virus-induced gene silencing (VIGS) is also shown. THS1 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 80; THS is a thebaine synthesis polypeptide isoform

represented by SEQ ID NO. 81; THS2 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 6, and THS2s is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 32. FIG. 12B show the primers used for the detection of specific THS genes and splice-variants and the analysis of VIGS experiments. (SEQ ID NOs. 65 to 78) The synthesized VIGS fragment consisting of two fused THS-specific transcript sequences is also provided. (SEQ ID NO. 79) FIG. 12C shows tissue-specific expression patterns of THS genes and splice-variants within *Papaver somniferum*. SYBR Green-based qRT-PCR was performed using gene-specific primers. Values represent the mean±standard deviation of three biological replicates.

(15) FIGS. 13A-13B. FIG. 13A demonstrates the detection of the pTRV2 (EV) and pTRV2-THS (THS) vectors in *Agrobacterium tumefaciens*-infiltrated plants. FIG. 13B shows virus-induced gene-silencing (VIGS) of THS transcripts and corresponding effects on selected alkaloid levels compared with empty-vector controls. Values represent the mean±standard deviation of three biological replicates.

(16) FIGS. 14A and 14B shows data from an in vitro enzyme assay of recombinant THS isoforms. THS1 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 80; THS is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 81; THS2 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 6, and THS2s is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 32. FIG. 14A shows affinity His.sub.6-tag purification of THS isoforms produced in *Escherichia coli*. Purified proteins were resolved on by SDS-PAGE. FIG. 14B shows in vitro enzyme assays of THS isoforms in a SalAT-coupled assay. THS1 and THS2 were active and increased thebaine (m/z 312) formation substantially at the expense of the m/z 330 compound(s). The THS1s and THS2s splice variants were not active.

(17) FIGS. 15A-15B shows homodimerization of THS proteins. THS1 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 80; THS1s is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 81; THS2 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 6, and THS2s is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 32. FIG. 15A is an SDS-PAGE and immunoblot analysis showing the cross-linking of purified recombinant THS1 and TH2 isoforms (asterisks in the middle near 46 kDa) and the formation of higher molecular weight oligomers (upper asterisks), and the limited cross-linking of the THS1s and TH2s splice-variants. FIG. 15B shows an SEC chromatography of purified recombinant THS2 showing a predominant homodimer with a molecular weight of ~40 kDa, and several higher molecular weight oligomers (upper asterisks).

(18) FIGS. 16A-16B shows substrate-competition assays for THS2 (a.k.a., thebaine synthesis polypeptide as encoded by SEQ ID NO. 6). FIG. 16A shows compounds tested as alternative substrates or competitive inhibitors of salutaridinol 7-O-acetate. FIG. 16B shows the relative conversion of salutaridinol 7-O-acetate to thebaine in standard THS2 assays using different concentrations of alternative substrates or competitive inhibitors.

(19) FIG. 17 shows relative thebaine titers produced from yeast. Bet v1 tested with HA epitope tag on C- or N-terminus (Bet v1-1-HA and HA-Bet v1-1, respectively) and expressed on a high copy plasmid. n=3. Bet v1 is also known as thebaine synthesis polypeptide or thebaine synthase.

(20) FIGS. 18A-18C. FIG. 18A displays the engineered yeast strains for production salutaridine and thebaine from norlaudanoline. FIG. 18B shows thebaine titers in supernatants of three engineered yeast strains (SC-2, SC-3 and SC-4). FIG. 18C shows relative peak area of unknown side product (m/z 330) in the supernatant in SC-3 and SC-4 strains. Yeast were grown in medium supplemented with 2.5 mM NLDS and 10 mM L-methionine for 96 hours. Error bars indicate standard deviations from four biological replicates.

(21) FIGS. 19A and 19B. FIG. 19A shows relative thebaine titers produced from yeast expressing various constructs. The various constructs include PR10-3; Bet v1-1; PR10-4; PR10-5; PR10-7; MLP15; MLP-2; and MLP-3, with either C- or N-terminus HA-tags. The N-terminally tagged Bet v1-1 (HA-Betv1-1) demonstrates the highest level of relative thebaine production. 50 μ M

salutaridine was fed for 48 hours. Each construct was expressed on a high copy plasmid. $n=3$. FIG. 19B shows the relative peak area of unknown side product (m/z 330). The N-terminally tagged Bet v1-1 (HA-Betv1-1) demonstrates the lowest level of relative m/z 330 production. Peak area is shown relative to the empty vector control. Error bars indicate standard deviations from three biological replicates.

(22) FIGS. 20A-20C. FIG. 20A shows relative thebaine titers produced from yeast expressing integrated Bet v1-1. 1 mM Salutaridine was fed for 48 hours. Bet v1-1 splice variants were tested from a single integrated copy on the genome (under a pGAL promoter). $N=4$. The control strain contained pGAL1 SalR and pTEA1 SalAT. Betv1-1 is a thebaine synthesis polypeptide represented by SEQ ID NO. 6 (i.e., Betv1-1 is the same as THS2). FIG. 20B shows relative thebaine titers produced from yeast expressing integrated Bet v1-1, Bet v1-1S, and Bet v1-1L. Bet v1-1S is a thebaine synthesis polypeptide represented by SEQ ID NO 32 (i.e., Betv1-1S is the same as THS2s). Bet v1-1L is a thebaine synthesis polypeptide represented by SEQ ID NO 31. Same conditions were used. FIG. 20C shows the amino acid sequences of the splice variants used in the experiments of FIG. 20B.

(23) FIGS. 21A-21C. FIG. 21A shows thebaine titers produced from three yeast strains (Sc-2, Sc-3, Sc-4, the components of which are shown in FIG. 18A). FIG. 21B shows reticuline titers produced from the three yeast strains. FIG. 21C shows salutaridine titers produced from the three yeast strains.

(24) FIGS. 22A-22C is the characterization of the predicted THS1 secretion signal peptide. THS1 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 80. FIG. 22A shows the effect of fusing green fluorescent protein (GFP) on the catalytic function of THS1 and THS2 in engineered yeast with chromosomally integrated SalR and SalAT. THS2 is a thebaine synthesis polypeptide isoform represented by SEQ ID NO. 6. Yeast were fed salutaridine and thebaine accumulation in the culture medium was monitored at 24 hours. FIG. 22B depicts the location of GFP fluorescence in various yeast strains containing a THS1-GFP and THS2-GFP fusions, compared with a native GFP, and fusion of GFP to a previously characterized protein possessing an N-terminal secretion signal (ssp-GFP). GFP-mediated fluorescence was only detected in cells, but not in the culture medium. FIG. 22C shows the cellular localization of GFP-mediated fluorescence in yeast producing native GFP, THS2-GFP and THS1-GFP. Only THS1, which possesses a putative secretion signal peptide, resulted in an altered localization of fluorescence associated the yeast cell periphery.

(25) FIGS. 23A-23D. FIGS. 23A-23C show titers of S-reticuline, R-reticuline, and salutaridine in identical yeast strains except for expressing 1 of 11 different purine permeases. The yeast strains were given a NLDS feed. FIG. 23 shows S-reticuline titers grown for 24 (left) and 48 (right) hours in the presence of an NLDS feed. FIG. 23B shows R-reticuline titers grown for 24 (left) and 48 (right) hours in the presence of an NLDS feed. FIG. 23C shows salutaridine titers grown for 24 (left) and 48 (right) hours in the presence of an NLDS feed. FIG. 23D shows NLDS and reticuline titers grown for 24 (left) and 48 (right) hours in the presence of an L-DOPA feed. PsomPUP1-4 was expressed on a high copy plasmid under the control of pGAL.

(26) FIG. 24 shows thebaine, salutaridinol, and salutaridine titers in the presence of one or more copies of purine permeases and Betv-1 (represented by SEQ ID NO. 6) after 48 hours. A nucleotide sequence encoding for BetV1 was integrated into the genome of the yeast strain. Further, a purine permease (PUP-1—SEQ ID NO 46)) was integrated into the genome of the yeast strain (control—left side). A PsomPUP1-4 (SEQ ID NO. 51 (amino acid) and 52 (nucleotide)) was expressed via a high copy plasmid (second from left). Additionally, JcurPUP1 (SEQ ID NO: 43 (amino acid) and 44 (nucleotide)) was integrated into the control strain, thus expressing both integrated copies of JcurPUP1 and PsomPUP1 (control—third from the left). Finally, three copies of a purine permease were expressed, PsomPUP1 integrated, JcurPUP1 integrated, and high copy plasmid expressing PUP1-4 (far right).

(27) FIG. 25 shows a graph showing concentrations of thebaine ($\mu\text{g/I/OD}$) present in growth medium comprising salutaridine following growth of different yeast strains expressing Betv-1 alone (HA_BetV1M and BetV1L_HA), or Betv-1 together with purine permease (HA_BetV1M-pp and BetV1L_HA-pp), as well as a control (EV). HA_BetV1M is an HA tagged thebaine synthesis polypeptide, where the thebaine synthesis polypeptide is represented by SEQ ID NO. 6. BetV1L_HA is an HA tagged thebaine synthesis polypeptide, where the thebaine synthesis polypeptide is represented by SEQ ID NO. 31. The purine permease used in this experiment is represented by SEQ ID NO. 35.

(28) FIG. 26 depicts a graph showing salutaridine ($\mu\text{g/I/OD}$) present in growth medium comprising salutaridine following growth of different yeast strains expressing Betv-1 alone (HA_BetV1M and BetV1L_HA), or Betv-1 together with purine permease (HA_BetV1M-pp and BetV1L_HA-pp), as well as a control (EV). HA_BetV1M is an HA tagged thebaine synthesis polypeptide, where the thebaine synthesis polypeptide is represented by SEQ ID NO. 6. BetV1L_HA is an HA tagged thebaine synthesis polypeptide, where the thebaine synthesis polypeptide is represented by SEQ ID NO. 31. The purine permease used in this experiment is represented by SEQ ID NO. 35.

(29) FIGS. 27A and 27B depicts a graph showing reticuline ($\mu\text{g/I/OD}$) present in a growth medium comprising either DOPA (FIG. 27A) or norlaudanosoline (NLDS) (FIG. 27B), following growth of yeast strains expressing DODC, MAO, NCS, 6OMT, CNMT and 4'OMT genes, and transformed with a gene expressing a first purine permease (PUP-L) or a second purine permease (PUP-N), each either alone or together with Betv-1. Betv-1 is a the thebaine synthesis polypeptide represented by SEQ ID NO. 6. The PUP-L is a purine permease represented by SEQ ID NO. 35. The PUP-N is a purine permease represented by SEQ ID NO. 37.

(30) FIGS. 28A and 28B depicts a graph showing reticuline and thebaine ($\mu\text{g/I/OD}$) present in a growth medium comprising (S)-reticuline in a yeast expressing REPI, CPR and SalSyn (FIG. 28A) or REPI, CPR, SalSyn, SalAT and SalR (FIG. 28B), each yeast strain transformed with a first purine permease (PUP-L) or a second purine permease (PUP-N), each either alone or together with Betv-1. Betv-1 is a the thebaine synthesis polypeptide represented by SEQ ID NO. 6. The PUP-L is a purine permease represented by SEQ ID NO. 35. The PUP-N is a purine permease represented by SEQ ID NO. 37.

(31) FIGS. 29A and 29B depicts a graph showing reticuline and thebaine ($\mu\text{g/I/OD}$) present in a growth medium comprising (R)-reticuline in a yeast expressing REPI, CPR and SalSyn (FIG. 29A) or REPI, CPR, SalSyn, SalAT and SalR (FIG. 29B), each yeast strain transformed with a first purine permease (PUP-L) or a second purine permease (PUP-N), each either alone or together with Betv-1. Betv-1 is a the thebaine synthesis polypeptide represented by SEQ ID NO. 6. The PUP-L is a purine permease represented by SEQ ID NO. 35. The PUP-N is a purine permease represented by SEQ ID NO. 37.

(32) FIGS. 30A and 30B depicts a graph showing thebaine ($\mu\text{g/I/OD}$) present in a growth medium comprising salutaridine in a yeast expressing SalR and SalAT (FIG. 30A) or REPI, CPR, SalSyn, SalAT and SalR (FIG. 30B), each yeast strain transformed with a first purine permease (PUP-L) or a second purine permease (PUP-N), each either alone or together with Betv-1. Betv-1 is a the thebaine synthesis polypeptide represented by SEQ ID NO. 6. The PUP-L is a purine permease represented by SEQ ID NO. 35. The PUP-N is a purine permease represented by SEQ ID NO. 37.

(33) The figures together with the following detailed description make apparent to those skilled in the art how the disclosure may be implemented in practice.

DETAILED DESCRIPTION OF THE DISCLOSURE

(34) BIAs can be produced in cells (e.g., microorganisms) by genetic engineering. For example, when producing morphinan alkaloids from microbial fermentation, a sugar substrate can be used to produce morphinan alkaloids. From sugar to the first morphinan alkaloid thebaine, there are at least 9 conversions required.

(35) In the first conversion, sugar, such as glucose, can be converted into L-tyrosine. FIG. 1A

shows the molecular pathway from glucose to L-tyrosine in yeast. In bacteria, glucose can be converted into L-tyrosine by using a variety of enzymes. FIG. 1B shows the molecular pathway from glucose to L-tyrosine in bacteria such as *E. coli*.

(36) The second conversion, 1-tyrosine to 1-DOPA can be performed by using a tyrosine hydroxylase (e.g., a cytochrome p450 such as CYP76AD1) to catalyze this reaction. The third conversion 1-DOPA to dopamine can be catalyzed by a DOPA decarboxylase (DDC). The fourth conversion makes use of one norcoclaurine synthase (NCS). The fifth conversion from (S)-norcoclaurine to (S)/(R)-reticuline takes advantage of one or more of several enzymes including but not limited to: 6OMT (6-O-Methyltransferase), CNMT (coclaurine N-methyltransferase), NMCH (cytochrome P450 N-methylcoclaurine hydroxylase a.k.a. CYP80B1), and 4OMT (4-O-Methyltransferase). The sixth conversion of (R)-reticuline to salutaridine requires CPR (cytochrome P450 reductase) and SAS (salutaridine synthase). The seventh conversion of salutaridine to salutaridinol uses salutaridine reductase. The eighth conversion of salutaridinol to salutaridinol-7-O-acetate takes advantage of salutaridinol-7-O-acetyltransferase. The ninth conversion of salutaridinol-7-O-acetate to thebaine was previously thought to be a spontaneous process (e.g., at an elevated pH). However, the inventors herein have found that a thebaine synthesis polypeptide can catalyze this reaction, orders of magnitude over a spontaneous reaction. Further, the addition of a purine permease can further greatly increase thebaine (and other intermediate) titers.

(37) Thebaine can be further converted into derivative morphinan alkaloids such as oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, and/or buprenorphine. For example, the conversion of thebaine to oripavine uses codeine O-demethylase (CODM). Oripavine can be further converted to morphinone using a thebaine 6-O-demethylase (T6ODM). Morphinone can be converted to morphine using codeinone reductase (COR). COR can also convert morphine into morphinone. Thebaine can be converted into neopinone by a thebaine 6-O-demethylase. The reaction of neopinone into codeinone is believed to be spontaneous. Codeinone can be converted into codeine through the use of COR. The reverse reaction from codeine to codeinone can also be catalyzed by COR. Codeine can be converted into morphine by using a CODM.

Definitions

(38) The terms “morphinan alkaloid”, or “morphinan alkaloid compound”, as may be used interchangeably herein, can refer to a class of chemical compounds comprising the prototype chemical structure shown in FIG. 2. Certain specific carbon and nitrogen atoms are numbered and may be referred to herein by reference to their position within the morphinan alkaloid structure e.g. C.sub.1, C.sub.2, N.sub.17, etc. The pertinent atom numbering is shown in FIG. 2.

(39) The term “oxo group”, as used herein, can refer to a group represented by “=O”.

(40) The term “O-acetyl group”, as used herein, can refer to a group represented by

(41) ##STR00001##

(42) The term “substrate morphinan alkaloid”, as used herein, can refer to a morphinan alkaloid compound that can be converted into a morphinan alkaloid such as thebaine or other morphinan alkaloid or derivatives thereof. For example, exogenously or endogenously provided substrate morphinan alkaloids can be more efficiently converted in a first in vivo or in vitro reaction mixture into thebaine in the presence of thebaine synthesis polypeptide, than in a second reaction mixture identical to the first reaction mixture but for the absence of thebaine synthesis polypeptide. A conversion is deemed “more efficient” if larger quantities of thebaine are obtained in the reaction mixture and/or if thebaine accumulates in the reaction mixture at a faster rate. Substrate morphinan alkaloids include, but are not limited to the morphinan alkaloid compounds set forth in FIGS. 3 A-C.

(43) The term “intermediate morphinan alkaloid compound”, as used herein, can refer to a morphinan alkaloid compound formed upon chemical conversion of a substrate morphinan alkaloid compound, and which subsequently can be converted into a BIA such as thebaine or other

morphinan alkaloid compound derivatives.

(44) The term “substrate”, as used herein, can refer to any compound that can be converted into a different compound. For example, 1-DOPA can be a substrate for DODC and can be converted into dopamine. For clarity, the term “substrate” includes all substrates including but not limited to base substrates, substrate morphinan alkaloids, and/or intermediate morphinan alkaloid compound.

(45) The term “base substrate”, as used herein, can refer to any exogenously provided organic carbon compounds which can be metabolized by a microorganism to permit the microorganism to produce an alkaloid, e.g., a morphinan alkaloid. Examples of base substrates include, without limitation, sugar compounds, such as, for example, glucose, fructose, galactose and mannose, as well as glycerol, L-tyrosine, tyramine, L-3,4-dihydroxyphenylalanine (L-DOPA), alcohols (e.g., ethanol), dopamine, or any combination thereof.

(46) The term “morphinan alkaloid derivative of thebaine”, as used herein can refer to a morphinan alkaloid compound that can be obtained by chemical conversion of thebaine and includes, without limitation, oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof.

(47) The term “thebaine synthesis polypeptide”, can refer to any polypeptide that can facilitate the conversion of a substrate into thebaine. For example, a thebaine synthesis polypeptide can be polypeptide that can convert salutaridinol-7-O-acetate into thebaine. In one example, thebaine synthesis polypeptide can be called “Bet v1” (or derivatives, fragments, variants thereof). Thebaine synthesis polypeptide can also be called “thebaine synthase” (THS). Further, variants/isoforms of the thebaine synthesis polypeptide are described throughout. For example, Betv1/Betv1M/BetV1-1/THS2 can refer to a sequence that is represented by SEQ ID NO. 6. BetV1L and BetV1-1L can refer to a sequence that is represented by SEQ ID NO. 31. THS2s and BetV1-1S can refer to a sequence that is represented by SEQ ID NO. 32. THS1 can refer to a sequence that is represented by SEQ ID NO. 80. THS1s can refer to a sequence that is represented by SEQ ID NO. 81. Bet v1-1A can refer to a C-terminal HA tagged Betv1, whereas Bet v1-1B can correspond to N-terminal HA tagged Betv1.

(48) The terms “salutaridinol 7-O-acetyltransferase”, “SalAT”, and “SalAT polypeptide”, can be used interchangeably herein, and can refer to any polypeptide that can facilitate the conversion of a substrate into salutaridinol 7-O-acetate. For example, SalAT can refer to any and all polypeptides that can convert salutaridinol to salutaridinol-7-O-acetate.

(49) The terms “salutaridine reductase”, “SalR”, and “SalR polypeptide”, can be used interchangeably herein, and can refer to any polypeptide that can facilitate the conversion of a substrate into salutaridinol. For example, SalR can refer to any and all polypeptides that can convert salutaridine to salutaridinol.

(50) The term “polynucleotide encoding a thebaine synthesis polypeptide” can refer to any and all polynucleotide sequences encoding a thebaine synthesis polypeptide.

(51) The terms “polynucleotide encoding a SalAT” and “polynucleotide encoding a SalAT polypeptide”, can be used interchangeably herein, and can refer to any and all polynucleotide sequences encoding a salutaridinol 7-O-acetyltransferase (“SalAT”) polypeptide.

(52) The terms “polynucleotide encoding a SalR” and “polynucleotide encoding a SalR polypeptide”, can be used interchangeably herein, and can refer to any and all polynucleotide sequences encoding a salutaridine reductase (“SalR”) polypeptide.

(53) The phrase “at least moderately stringent hybridization conditions” can mean that conditions are selected which promote selective hybridization between two complementary polynucleotide molecules in solution. Hybridization may occur to all or a portion of a polynucleotide molecule. The hybridizing portion is typically at least 15 (e.g. 20, 25, 30, 40 or 50) nucleotides in length. Those skilled in the art will recognize that the stability of a polynucleotide duplex, or hybrids, is determined by the T_m , which in sodium containing buffers is a function of the sodium ion concentration and temperature ($T_m = 81.5^\circ \text{C} + 16.6 (\log_{10} [\text{Na}^+]) + 0.41 (\% (\text{G} + \text{C})) / 600 / 1$), or

similar equation). Accordingly, the parameters in the wash conditions that determine hybrid stability are sodium ion concentration and temperature. In order to identify molecules that are similar, but not identical, to a known polynucleotide molecule a 1% mismatch may be assumed to result in about a 1° C. decrease in T_m, for example if polynucleotide molecules are sought that have a >95% identity, the final wash temperature will be reduced by about 5° C. Based on these considerations those skilled in the art will be able to readily select appropriate hybridization conditions. In preferred embodiments, stringent hybridization conditions are selected. By way of example the following conditions may be employed to achieve stringent hybridization: hybridization at 5× sodium chloride/sodium citrate (SSC)/5×Denhardt's solution/1.0% SDS at T_m (based on the above equation) −5° C., followed by a wash of 0.2×SSC/0.1% SDS at 60° C. Moderately stringent hybridization conditions include a washing step in 3×SSC at 42° C. It is understood however that equivalent stringencies may be achieved using alternative buffers, salts and temperatures. Additional guidance regarding hybridization conditions may be found in: Current Protocols in Molecular Biology, John Wiley & Sons, N.Y., 1989, 6.3.1.-6.3.6 and in: Sambrook et al., Molecular Cloning, a Laboratory Manual, Cold Spring Harbor Laboratory Press, 1989, Vol. 3.

(54) The term “substantially identical” and its grammatical equivalents in reference to another sequence as used herein can mean at least 50% identical. In some instances, the term substantially identical refers to a sequence that is 55% identical. In some instances, the term substantially identical refers to a sequence that is 60% identical. In some instances, the term substantially identical refers to a sequence that is 65% identical. In some instances, the term substantially identical refers to a sequence that is 70% identical. In some instances, the term substantially identical refers to a sequence that is 75% identical. In some instances, the term substantially identical refers to a sequence that is 80% identical. In other instances, the term substantially identical refers to a sequence that is 81% identical. In other instances, the term substantially identical refers to a sequence that is 82% identical. In other instances, the term substantially identical refers to a sequence that is 83% identical. In other instances, the term substantially identical refers to a sequence that is 84% identical. In other instances, the term substantially identical refers to a sequence that is 85% identical. In other instances, the term substantially identical refers to a sequence that is 86% identical. In other instances, the term substantially identical refers to a sequence that is 87% identical. In other instances, the term substantially identical refers to a sequence that is 88% identical. In other instances, the term substantially identical refers to a sequence that is 89% identical. In some instances, the term substantially identical refers to a sequence that is 90% identical. In some instances, the term substantially identical refers to a sequence that is 91% identical. In some instances, the term substantially identical refers to a sequence that is 92% identical. In some instances, the term substantially identical refers to a sequence that is 93% identical. In some instances, the term substantially identical refers to a sequence that is 94% identical. In some instances, the term substantially identical refers to a sequence that is 95% identical. In some instances, the term substantially identical refers to a sequence that is 96% identical. In some instances, the term substantially identical refers to a sequence that is 97% identical. In some instances, the term substantially identical refers to a sequence that is 98% identical. In some instances, the term substantially identical refers to a sequence that is 99% identical. In some instances, the term substantially identical refers to a sequence that is 100% identical. In order to determine the percentage of identity between two sequences, the two sequences are aligned, using for example the alignment method of Needleman and Wunsch (J. Mol. Biol., 1970, 48: 443), as revised by Smith and Waterman (Adv. Appl. Math., 1981, 2: 482) so that the highest order match is obtained between the two sequences and the number of identical amino acids/nucleotides is determined between the two sequences. For example, methods to calculate the percentage identity between two amino acid sequences are generally art recognized and include, for example, those described by Carillo and Lipton (SIAM J. Applied Math., 1988, 48:1073) and those described in Computational Molecular

Biology, Lesk, e.d. Oxford University Press, New York, 1988, Biocomputing: Informatics and Genomics Projects. Generally, computer programs will be employed for such calculations. Computer programs that may be used in this regard include, but are not limited to, GCG (Devereux et al., *Nucleic Acids Res.*, 1984, 12: 387) BLASTP, BLASTN and FASTA (Altschul et al., *J. Molec. Biol.*, 1990:215:403). A particularly preferred method for determining the percentage identity between two polypeptides involves the Clustal W algorithm (Thompson, J D, Higgins, D G and Gibson T J, 1994, *Nucleic Acid Res* 22(22): 4673-4680 together with the BLOSUM 62 scoring matrix (Henikoff S & Henikoff, J G, 1992, *Proc. Natl. Acad. Sci. USA* 89: 10915-10919 using a gap opening penalty of 10 and a gap extension penalty of 0.1, so that the highest order match obtained between two sequences wherein at least 50% of the total length of one of the two sequences is involved in the alignment.

(55) The term “chimeric”, as used herein in the context of polynucleotides, can refer to at least two polynucleotides that are not naturally linked. Chimeric polynucleotides include linked polynucleotides of different natural origins. For example, a polynucleotide constituting a yeast promoter linked to a polynucleotide sequence encoding a plant thebaine synthesis polypeptide is considered chimeric. Chimeric polynucleotides also may comprise polynucleotides of the same natural origin, provided they are not naturally linked. For example a polynucleotide sequence constituting a promoter obtained from a particular cell-type may be linked to a polynucleotide sequence encoding a polypeptide obtained from that same cell-type, but not normally linked to the polynucleotide sequence constituting the promoter. Chimeric polynucleotide sequences also include polynucleotide sequences comprising naturally occurring polynucleotide sequences linked to non-naturally occurring polynucleotide sequences.

(56) The term “isolated”, as used herein, describes a compound, e.g., a morphinan alkaloid or a polypeptide, which has been separated from components that naturally accompany it. Typically, a compound is isolated when at least 60%, 75%, 80%, 85%, 90%, 95%, 96%, 97%, 98%, or 99% of the total material (by volume, by wet or dry weight, or by mole percent or mole fraction) in a sample is the compound of interest. The percentage of a material of interest can be measured by any appropriate method, e.g., in the case of polypeptides, by chromatography, gel electrophoresis or HPLC analysis.

(57) The term “heterologous” and its grammatical equivalents as used herein can mean “derived from a different species or cell type.” For example, a “heterologous gene” can mean a gene that is from a different species or cell type. For example, a heterologous thebaine synthesis polypeptide can refer to a thebaine synthesis polypeptide which is present in a cell or cell type in which it is not naturally present.

(58) The term “functional variant”, as used herein, in reference to polynucleotide sequences or polypeptide sequences, can refer to polynucleotide sequences or polypeptide sequences capable of performing the same function as a noted polynucleotide sequence or polypeptide sequence. It is contemplated that for any of the polypeptides disclosed throughout, a functional variant can also be used in addition to the polypeptide or the polypeptide can be substituted.

(59) The term “in vivo”, as used herein, can refer to within a living cell, including, for example, a microorganism or a plant cell.

(60) The term “in vitro”, as used herein, can refer to outside a living cell, including, without limitation, for example, in a microwell plate, a tube, a flask, a beaker, a tank, a reactor and the like.

(61) It should be noted that terms of degree such as “substantially”, “about” and “approximately”, as used herein, can refer to a reasonable amount of deviation of the modified term such that the end result is not significantly changed. These terms of degree should be construed as including a deviation of the modified term if this deviation would not negate the meaning of the term it modifies. For example, in relation to a reference numerical value the terms of degree can include a range of values plus or minus 10% from that value. For example, with regards to “about”, the amount “about 10” can include any amounts from 9 to 11. The terms of degree in relation to a

reference numerical value can also include a range of values plus or minus 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, or 1% from that value.

(62) As used herein, the wording “and/or” is intended to represent an inclusive-or. That is, “X and/or Y” is intended to mean X or Y or both, for example. As a further example, “X, Y, and/or Z” is intended to mean X or Y or Z or any combination thereof.

(63) As used herein, the word “assist” and as used herein can mean to help, support, facilitate, or the like. For example, when used in context with an enzyme that converts X to Y, a polypeptide that “assists” the enzyme can increase the efficiency of which X is converted to Y. In some cases, the polypeptide that “assists” cannot directly convert X into Y.

(64) General Implementation

(65) Certain alkaloids belong to a class of chemical compounds known as benzyloquinoline alkaloids (“BIAs”). Certain polypeptides capable of mediating chemical reactions involving the conversion of a substrate (e.g., a substrate benzyloquinoline) into a product (e.g., a product benzyloquinoline) are disclosed. The benzyloquinoline (either the substrate or product) can belong to the benzyloquinoline subclass of morphinan alkaloids, depending on what product is desired. Accordingly, disclosed are certain polypeptides capable of mediating chemical reactions involving conversion of a substrate into thebaine or other target morphinan alkaloids or morphinan alkaloid derivatives. Further, disclosed are methods that represent a novel and efficient means of making thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives.

(66) The general pathways from a sugar substrate to a BIA, such as thebaine, includes enzymes that can perform any one of the following reactions: i) sugar to 1-tyrosine; ii) 1-tyrosine to 1-DOPA; iii) 1-DOPA to dopamine; iv) dopamine to (S)-norcoclaurine; v) (S)-norcoclaurine to (S)/(R)-reticuline; vi) (R)-reticuline to salutaridine; vii) salutaridine to salutaridinol; viii) salutaridinol to salutaridinol-7-O-acetate; ix) salutaridinol-7-O-acetate to the BIA thebaine; x) thebaine to other BIAs (such as oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, and/or buprenorphine). Depending on the substrate used, any of the products of the pathway can be made by increasing or decreasing the expression of the enzymes that promote the formation of the desired product.

(67) Some of the chemical conversions can be performed by one or more enzymes, including but not limited to tyrosine hydroxylase (TYR); DOPA decarboxylase (DODC); norcoclaurine synthase (NCS); 6-O-Methyltransferase (6OMT); coclaurine N-methyltransferase (CNMT), cytochrome P450 N-methylcoclaurine hydroxylase (NMCH), and 4-O-methyltransferase (4OMT); cytochrome P450 reductase (CPR), salutaridine synthase (SAS); salutaridine reductase (SalR); salutaridinol-7-O-acetyltransferase (SalAT); purine permease (PUP); or any combination thereof. Further to form a BIA, such as thebaine or morphinan alkaloids, the product should be contacted with a thebaine synthesis polypeptide; a codeine O-demethylase (CODM); a thebaine 6-O-demethylase (T6ODM); a codeinone reductase (COR); or any combination thereof.

(68) Some of the methods herein involve the use of novel polypeptides known as thebaine synthesis polypeptides, as well as novel polynucleotides encoding the thebaine synthesis polypeptides. The thebaine synthesis polypeptides disclosed and used throughout, can be a polypeptide that is capable of converting salutaridinol-7-O-acetate to thebaine. Currently, it is thought that the reaction of converting salutaridinol-7-O-acetate to thebaine occurs by a spontaneous reaction at certain elevated pH levels. However, this reaction is extremely slow and thermodynamically unfavorable at normal conditions. Some of methods described throughout take advantage of using a thebaine synthesis polypeptide that can convert the salutaridinol-7-O-acetate to thebaine at an accelerated rate.

(69) Some of the methods herein involve the use of a purine permease. In some cases, the purine permease can be used in combination with other polypeptides, such as a thebaine synthesis polypeptide (THS). The purine permease disclosed and used throughout, can be a polypeptide that is capable of increase the production titers of thebaine or other BIA intermediate by any cell or

methods disclosed throughout.

(70) Different substrates can be used with the enzymes described throughout. For example, one such substrate can be a substrate morphinan alkaloid having the chemical formula (I):

(71) ##STR00002## wherein, wherein R.sub.1 represents a hydrogen atom and R.sub.1' represents an O-acetyl group, or wherein R.sub.1 represents a hydrogen atom and R.sub.1' represents a hydroxyl group, or wherein, taken together, R.sub.1 and R.sub.1' form an oxo group.

(72) The substrate can also have the chemical formula (II):

(73) ##STR00003## wherein, wherein R.sub.1 represents a hydrogen atom and R.sub.1' represents an O-acetyl group, or wherein R.sub.1 represents a hydrogen atom and R.sub.1' represents a hydroxyl group, or wherein, taken together, R.sub.1 and R.sub.1' form an oxo group.

(74) The substrate can also have the chemical formula (III):

(75) ##STR00004## wherein, wherein R.sub.1 represents a hydrogen atom and R.sub.1' represents an O-acetyl group, or wherein R.sub.1 represents a hydrogen atom and R.sub.1' represents a hydroxyl group, or wherein, taken together, R.sub.1 and R.sub.1' form an oxo group.

(76) The substrate can also have the chemical formula (IV):

(77) ##STR00005## wherein, wherein R.sub.1 represents a hydrogen atom and R.sub.1' represents an O-acetyl group, or wherein R.sub.1 represents a hydrogen atom and R.sub.1' represents a hydroxyl group, or wherein, taken together, R.sub.1 and R.sub.1' form an oxo group.

(78) In embodiments where substrates are used, for example substrates that have the chemical formula represented by (I), (II), (III), or (IV), one or more additional polypeptides can assist in the reaction. For example, SalAT polypeptide and a SalR polypeptide can be used to make a BIA, such as thebaine. Other polypeptides can assist in making thebaine as well. However, focusing here on SalAT and SalR, the SalAT polypeptide and/or a SalR polypeptide can be provided through adding the polypeptides directly to the culture media, transfections (transient or stable) of polynucleotide into the cells prior to culturing, or other means.

(79) Synthesis of Benzylisoquinoline Alkaloids (Bias)

(80) In Vivo Synthesis of BIAs

(81) BIAs (e.g., thebaine, oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, and/or buprenorphine) can be produced at least in two ways. The first way is that the BIA is synthesized within a cell. This is referred to as the in vivo synthesis of the BIAs. This generally requires the use of a cell that can naturally produce BIAs from a substrate, or the use of a genetically modified cell that is either 1) enhanced to produce more BIAs compared to a non-modified cell or 2) produces BIAs even though it does not do so naturally.

(82) Genetically Modified Cells to Produce BIAs

(83) Thebaine Synthesis Polypeptide

(84) Described herein is a genetically modified cell (e.g., a microorganism) that comprises a polynucleotide encoding a heterologous enzyme capable of converting salutaridinol-7-O-acetate to thebaine. The enzyme that is capable of converted salutaridinol-7-O-acetate to thebaine can be a thebaine synthesis polypeptide. Currently, there is no known enzyme that converts salutaridinol-7-O-acetate to thebaine, as it is thought that this conversion occurs spontaneously.

(85) The thebaine synthesis polypeptide can be artificially synthesized or altered from how it exists in nature.

(86) The thebaine synthesis polypeptide can also be expressed in cells that it is not normally expressed in. For example, a thebaine synthesis polypeptide from a plant can be expressed in yeast. This generally requires that the thebaine synthesis polypeptide be under the control of a yeast promoter.

(87) The thebaine synthesis polypeptide can be from a plant. For example, the thebaine synthesis polypeptide can be from a plant that is from the genus *Papaver*. More specifically, *Papaver* plants that can be used include, but are not limited to *Papaver bracteatum*, *Papaver somniferum*, *Papaver*

cylindricum, *Papaver decaisnei*, *Papaver fugax*, *Papaver nudicale*, *Papaver oreophyllum*, *Papaver orientate*, *Papaver paeonifolium*, *Papaver persicum*, *Papaver pseudo-orientale*, *Papaver rhoeas*, *Papaver rhopalotheca*, *Papaver armeniacum*, *Papaver setigerum*, *Papaver tauricolum*, and *Papaver triniaefolium*. A particularly useful *Papaver* thebaine synthesis polypeptide is a *Papaver somniferum* thebaine synthesis polypeptide.

(88) The thebaine synthesis polypeptide can be a polypeptide that is encoded by a polynucleotide that is substantially identical to SEQ ID NO. 19. For example, the thebaine synthesis polypeptide can be encoded by a polynucleotide that is at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% identical to any one of SEQ. ID NO. 19. Further, codon optimized polynucleotides (for a particular host cell/organism) for the above referenced sequences can be used herein.

(89) In some cases, the thebaine synthesis polypeptide can be encoded by a polynucleotide capable of hybridizing under at least moderately stringent conditions to a polynucleotide that is substantially identical to SEQ ID NO. 19 or codon optimized sequenced.

(90) The thebaine synthesis polypeptide can also be a polypeptide having an amino acid sequence that is substantially identical to SEQ ID NO. 6. For example, the thebaine synthesis polypeptide can be an amino acid sequence that is at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% identical to any one of SEQ ID NO. 6. Polypeptides that are substantially identical to SEQ ID NO. 6 can be used as a thebaine synthesis polypeptide.

(91) In some cases, the thebaine synthesis polypeptide can be a fragment thereof. The fragment can still retain thebaine synthesis activity. In some cases, the thebaine synthesis activity can be decreased or increased compared to the activity produced by SEQ ID NO. 6. In one case, the thebaine synthesis polypeptide can a polypeptide having an amino acid sequence that is substantially identical to SEQ ID NO. 31 or 32.

(92) In some cases, the thebaine synthesis polypeptide can be a variant or isoform. In some cases, variant or isoform can still retain thebaine synthesis activity. In some cases, the thebaine synthesis activity can be decreased or increased compared to the activity produced by SEQ ID NO. 6. In one case, the thebaine synthesis polypeptide can a polypeptide having an amino acid sequence that is substantially identical to SEQ ID NO. 80 or 81.

(93) In some cases, the thebaine synthesis polypeptide can comprise a sequence encoding for a secretion signal, HA-tag, myc-tag, or other signal/tag. In some cases, these tags or signals do not affect the activity of the thebaine synthesis polypeptide.

(94) Polypeptides that can Assist the Thebaine Synthesis Polypeptide

(95) The thebaine synthesis polypeptide can also be assisted by one or more polypeptides. For example, thebaine synthesis polypeptide can be assisted (e.g., help synthesize more thebaine) by one or more polypeptides that are encoded by a polynucleotide that is substantially identical to SEQ ID NO. 18; SEQ ID NO. 20; SEQ ID NO. 21; SEQ ID NO. 22; SEQ ID NO. 23; SEQ ID NO. 24; SEQ ID NO. 25; SEQ ID NO. 26; SEQ ID NO. 27; SEQ ID NO. 28; SEQ ID NO. 29 or SEQ ID NO. 30. Codon optimized polynucleotides (for a particular host cell/organism) for the above referenced sequences can be used herein.

(96) In some cases, the thebaine synthesis polypeptide can also be assisted by one or more polypeptides that are capable of hybridizing under at least moderately stringent conditions to a polynucleotide that is substantially identical to SEQ ID NO. 18; SEQ ID NO. 20; SEQ ID NO. 21; SEQ ID NO. 22; SEQ ID NO. 23; SEQ ID NO. 24; SEQ ID NO. 25; SEQ ID NO. 26; SEQ ID NO. 27; SEQ ID NO. 28; SEQ ID NO. 29 or SEQ ID NO. 30, or codon optimized sequences.

(97) In some instances the thebaine synthesis polypeptide can also be assisted by one or more polypeptides that is substantially identical to SEQ ID NO. 5; SEQ ID NO. 7; SEQ ID NO. 8; SEQ ID NO. 9; SEQ ID NO. 10; SEQ ID NO. 11; SEQ ID NO. 12; SEQ ID NO. 13; SEQ ID NO. 14; SEQ ID NO. 15; SEQ ID NO. 16 or SEQ ID NO. 17.

(98) For example, the thebaine synthesis polypeptide can also be assisted by one or more

polypeptides that is at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% identical to any one of SEQ. ID NOs 5, and 7 to 17. The thebaine synthesis polypeptide can also be assisted by one or more polypeptides that is substantially identical to SEQ ID NO. 5. The thebaine synthesis polypeptide can be assisted by one or more polypeptides that is substantially identical to SEQ ID NO. 7 can also be used. The thebaine synthesis polypeptide can be assisted by one or more polypeptides that is substantially identical to SEQ ID NO. 8. Polypeptides that are substantially identical to SEQ ID NO. 9 can be used to assist the thebaine synthesis polypeptide. A polypeptide that is substantially identical to SEQ ID NO. 10 can also be used to assist thebaine synthesis polypeptide. The thebaine synthesis polypeptide can be assisted by one or more polypeptides that is substantially identical to SEQ ID NO. 11. Polypeptides that are substantially identical to SEQ ID NO. 12 can be used to assist thebaine synthesis polypeptide. A polypeptide that is substantially identical to SEQ ID NO. 13 can also be used to assist a thebaine synthesis polypeptide. The thebaine synthesis polypeptide can be assisted by one or more polypeptides that is substantially identical to SEQ ID NO. 14. Polypeptides that are substantially identical to SEQ ID NO. 15 can be used to assist thebaine synthesis polypeptide. A polypeptide that is substantially identical to SEQ ID NO. 16 can also be used to assist a thebaine synthesis polypeptide. The thebaine synthesis polypeptide can be assisted by one or more polypeptides that is substantially identical to SEQ ID NO. 17.

(99) Purine Permease

(100) Described herein is a genetically modified cell (e.g., a microorganism) that comprises a polynucleotide encoding an enzyme (e.g., heterologous enzyme) capable of increasing thebaine titers compared to a non-genetically modified cell. The enzyme that is capable of increasing thebaine titers can be a purine permease. The purine permease can sometimes work by itself or in conjunction with a thebaine synthesis polypeptide or other enzyme. For example, in some cases, the combination of purine permease and thebaine synthesis polypeptide can significantly increase thebaine titers compared to a non-genetically modified cell or a cells that is transformed with only a purine permease or only a thebaine synthesis polypeptide.

(101) The purine permease can be artificially synthesized or altered from how it exists in nature.

(102) The purine permease can also be expressed in cells that it is not normally expressed in. For example, a purine permease from a plant can be expressed in yeast. This generally requires that the purine permease be under the control of a yeast promoter.

(103) The purine permease can be from a plant. For example, the purine permease can be from a plant that is from the genus *Papaver*. More specifically, *Papaver* plants that can be used include, but are not limited to *Papaver bracteatum*, *Papaver somniferum*, *Papaver cylindricum*, *Papaver decaisnei*, *Papaver fugax*, *Papaver nudicale*, *Papaver oreophyllum*, *Papaver orientate*, *Papaver paeonifolium*, *Papaver persicum*, *Papaver pseudo-orientale*, *Papaver rhoeas*, *Papaver rhopalotheca*, *Papaver armeniacum*, *Papaver setigerum*, *Papaver tauricolum*, and *Papaver triniaefolium*. A particularly useful purine permease is a *Papaver somniferum* purine permease.

(104) The purine permease can be encoded by a polynucleotide that is substantially identical to SEQ ID NO. 36. For example, the purine permease can be encoded by a polynucleotide that is at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% identical to any one of SEQ. ID NO. 36. Further, codon optimized polynucleotides (for a particular host cell/organism) for the above referenced sequences can be used herein.

(105) In some cases, the purine permease can be encoded by a polynucleotide capable of hybridizing under at least moderately stringent conditions to a polynucleotide that is substantially identical to SEQ ID NO. 36 or codon optimized sequenced. Additionally, the purine permease can be substantially identical to SEQ ID NO. 38 or 39.

(106) The purine permease can also be a polypeptide having an amino acid sequence that is substantially identical to SEQ ID NO. 35. For example, the purine permease can be an amino acid sequence that is at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% identical

to any one of SEQ ID NO. 35. Polypeptides that are substantially identical to SEQ ID NO. 35 can be used as a purine permease. The purine permease can also be a polypeptide having an amino acid sequence that is substantially identical to SEQ ID NO. 37. For example, the purine permease can be an amino acid sequence that is at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% identical to any one of SEQ ID NO. 37. Polypeptides that are substantially identical to SEQ ID NO. 37 can be used as a purine permease.

(107) In some cases, the purine permease can be encoded by a polynucleotide that is substantially identical to any one of SEQ ID NOs. 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, or 64. For example, the purine permease can be encoded by a polynucleotide that is at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% identical to any one of SEQ ID NOs. 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, or 64. Further, codon optimized polynucleotides (for a particular host cell/organism) for the above referenced sequences can be used herein.

(108) In some cases, the purine permease can be encoded by a polynucleotide capable of hybridizing under at least moderately stringent conditions to a polynucleotide that is substantially identical to any one of SEQ ID NOs. 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, or 64 or codon optimized sequences.

(109) The purine permease can also be a polypeptide having an amino acid sequence that is substantially identical to any one of SEQ ID NOs. 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, or 63. For example, the purine permease can be an amino acid sequence that is at least 50%, 60%, 70%, 80%, 85%, 90%, 95%, 96%, 97%, 98% or 99% identical to any one of SEQ ID NOs. 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, or 63. Polypeptides that are substantially identical to SEQ ID NOs. 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, or 63 can be used as a purine permease.

(110) In some cases, the purine permease can be a fragment thereof. The fragment can still retain purine permease activity. In some cases, the fragment can result in purine permease activity that is decreased or increased compared to the activity produced by any one of SEQ ID NO. 35, 37, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, or 63.

(111) Additional Enzymes Useful in Making BIAS

(112) The genetically modified cell can also comprise a polynucleotide encoding a SalAT polypeptide and/or a SalR polypeptide. The SalAT and/or SalR can be from a plant, for example, a plant from the genus *Papaver*, or from *Papaver* species such as *Papaver bracteatum*, *Papaver somniferum*, *Papaver cylindricum*, *Papaver decaisnei*, *Papaver fugax*, *Papaver nudicale*, *Papaver oreophyllum*, *Papaver orientale*, *Papaver paeonifolium*, *Papaver persicum*, *Papaver pseudo-orientale*, *Papaver rhoeas*, *Papaver rhopalotheca*, *Papaver armeniacum*, *Papaver setigerum*, *Papaver tauricolum*, and *Papaver triniaefolium*. A particularly useful *Papaver* SalAT and/or SalR is a *Papaver somniferum* SalAT and/or SalR.

(113) In certain instances, the SalAT polypeptide can be encoded by a polynucleotide sequence that is substantially identical to SEQ ID NO. 1 or a fragment or functional variant thereof. Additionally, the SalAT can in some instances be encoded by a polynucleotide capable of hybridizing under at least moderately stringent conditions to a polynucleotide (e.g., SEQ ID NO. 1) encoding a SalAT polypeptide. In some cases, the SalAT can comprise an amino acid sequence which is substantially identical to SEQ ID NO. 2.

(114) The SalR polypeptide can be encoded by a polynucleotide sequence that is substantially identical to SEQ ID NO. 3 or a fragment or functional variant thereof. Additionally, the SalR can in some cases be encoded by a polynucleotide capable of hybridizing under at least moderately stringent conditions to any polynucleotide (e.g., SEQ ID NO. 3) encoding a SalR polypeptide. In some cases, the SalR polypeptide can comprise an amino acid sequence which is substantially identical to SEQ ID NO. 4.

(115) The genetically modified cell can also further comprises one or more nucleic acids encoding for an enzyme capable of catalyzing one or more of the reactions: a) a sugar to L-tyrosine; b) L-tyrosine to L-DOPA; c) L-DOPA to Dopamine; d) Dopamine to (S)-Norcoclaurine; e) (S)-

Norcoclaurine to (S)/(R)-Reticuline; f) (R)-Reticuline to Salutaridine; g) Salutaridine to Salutaridinol; h) Salutaridinol to Salutaridinol-7-O-acetate; or i) thebaine to oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof. Depending on the substrate used, any of the products of the pathway can be made by increasing or decreasing the expression of the enzymes that promote the formation of the desired product.

(116) The genetically modified cell can also further one or more enzymes (in some cases heterologous enzymes), including but not limited to tyrosine hydroxylase (TYR); DOPA decarboxylase (DODC); norcoclaurine synthase (NCS); 6-O-Methyltransferase (6OMT); coclaurine N-methyltransferase (CNMT), cytochrome P450 N-methylcoclaurine hydroxylase (NMCH), and 4-O-methyltransferase (4OMT); cytochrome P450 reductase (CPR), salutaridine synthase (SAS); salutaridine reductase (SalR); salutaridinol-7-O-acetyltransferase (SalAT); purine permease (PUP); or any combination thereof. Further to form a BIA, such as thebaine or morphinan alkaloids, contacting the product of should be contacted with a thebaine synthesis polypeptide; a codeine O-demethylase (CODM); a thebaine 6-O-demethylase (T6ODM); a codeinone reductase (COR); or any combination thereof.

(117) The expression (or overexpression) of the thebaine synthesis polypeptide in certain cells can lead to increased production of BIAs, such as thebaine. Additionally, the expression of other enzymes within the sugar.fwdarw.BIA pathway, can result in greater production of BIAs, such as thebaine. For example, expression (or overexpression) of purine permease in certain cells can lead to increased production of BIAs, such as thebaine.

(118) Fragments, including functional fragments, of any and all polypeptides described here, are contemplated herein, even without directly mentioning fragments of the polypeptides. For example, fragments of thebaine synthesis polypeptides and/or purine permease are contemplated.

(119) Polynucleotides Useful in Making BIAs

(120) The polynucleotide encoding the thebaine synthetic polypeptide or any enzyme disclosed herein can be linked to a polynucleotide capable of controlling expression of the thebaine synthesis polypeptide and/or purine permease (or other enzyme) in a cell. A chimeric polynucleotide comprising as operably linked components: (a) one or more polynucleotides capable of controlling expression in a host cell; and (b) a polynucleotide encoding a thebaine synthetic polypeptide and/or purine permease (or other enzymes) can be used to express the desired polypeptide/enzyme.

(121) Polynucleotides capable of controlling expression in host cells that may be used herein include any transcriptional promoter capable of controlling expression of polypeptides in host cells. Generally, promoters obtained from bacterial cells are used when a bacterial host is selected in accordance herewith, while a fungal promoter will be used when a fungal host is selected, a plant promoter will be used when a plant cell is selected, and so on. Further polynucleotide elements capable of controlling expression in a host cell include transcriptional terminators, enhancers and the like, all of which may be included in the polynucleotides molecules (including chimeric polynucleotides) disclosed herein. A skilled artisan would understand that operable linkage of polynucleotide sequences can include linkage of promoters and sequences capable of controlling expression to coding sequences in the 5' to 3' direction of transcription.

(122) The polynucleotide molecules disclosed herein can be integrated into a recombinant expression vector which ensures good expression in the host cell. For example, the polynucleotides herein can comprise one or more polynucleotides capable of controlling expression in a host cell and can subsequently be linked to a polynucleotide sequence encoding a thebaine synthetic polypeptide and/or purine permease (or other sequence encoding for a desired enzyme). These vectors can be stably integrated into the genome of a desired cell or simply transfected without any integration.

(123) Disclosed is also a recombinant polynucleotide expression vector comprising as operably linked components: (a) one or more polynucleotides capable of controlling expression in a host

cell; and (b) a polynucleotide encoding a thebaine synthetic polypeptide and/or purine permease, wherein the expression vector is suitable for expression in a host cell. The term "suitable for expression in a host cell" can mean that the recombinant polynucleotide expression vector comprises the chimeric polynucleotide molecule linked to genetic elements required to achieve expression in a host cell.

(124) Genetic elements that can be included in the expression vector include a transcriptional termination region, one or more polynucleotides encoding marker genes, one or more origins of replication and the like. The expression vector can also further comprise genetic elements required for the integration of the vector or a portion thereof in the host cell's genome. For example, if a plant host cell is used the T-DNA left and right border sequences which facilitate the integration into the plant's nuclear genome can be used.

(125) The recombinant expression vector can further contain a marker gene. Marker genes that can be used and included in all genes that allow the distinction of transformed cells from non-transformed cells, including selectable and screenable marker genes. A marker gene may be a resistance marker such as an antibiotic resistance marker against, for example, kanamycin or ampicillin. Screenable markers that may be employed to identify transformants through visual inspection include β -glucuronidase (GUS) (see e.g., U.S. Pat. Nos. 5,268,463 and 5,599,670) and green fluorescent protein (GFP) (see e.g., Niedz et al., 1995, Plant Cell Rep., 14: 403).

(126) The preparation of the vectors may be accomplished using commonly known techniques such as restriction digestion, ligation, gel electrophoresis, DNA sequencing, Polymerase Chain Reaction (PCR) and other methodologies. A wide variety of cloning vectors are available to perform the necessary steps required to prepare a recombinant expression vector. Among the vectors with a replication system functional in bacteria such as *E. coli* are vectors such as pBR322, the pUC series of vectors, the M13 mp series of vectors, pBluescript etc. Typically these cloning vectors contain a marker allowing selection of transformed cells. Polynucleotides may be introduced in these vectors, and the vectors may be introduced in the host cell by preparing competent cells, electroporation or using other methodologies. The host cell may be grown in an appropriate medium including but not limited to, Luria-Broth medium, dulbecco eagle modified medium (DMEM) or Opti-mem, and subsequently harvested. Recombinant expression vectors can be recovered from cells upon harvesting and lysing of the cells. Further, general guidance with respect to the preparation of recombinant vectors and growth of recombinant organisms may be found in, for example: Sambrook et al., Molecular Cloning, a Laboratory Manual, Cold Spring Harbor Laboratory Press, 2001, Third Ed.

(127) Modifying Endogenous Gene Expression

(128) The genetically modified cells disclosed herein can have their endogenous genes regulated. This can be useful, for example, when there is negative feedback to the expression of a desired polypeptide, such as a thebaine synthesis polypeptide and/or purine permease. Modifying this negative regulator can lead to increased expression of a desired polypeptide.

(129) Therefore, disclosed herein is a method for modifying expression of polynucleotide sequences in a cell naturally expressing a desired polypeptide, the method comprising: (a) providing a cell naturally expressing a desired polypeptide; (b) mutagenizing the cell; (c) growing the cell to obtain a plurality of cells; and (d) determining if the plurality of cells comprises a cell with modulated levels of the desired polypeptide. In some cases, the desired polypeptide is a thebaine synthesis polypeptide. In some cases, the desired polypeptide is a purine permease. In other cases, the desired polypeptide is an enzyme that can perform any one of the following reactions: i) sugar to 1-tyrosine; ii) 1-tyrosine to 1-DOPA; iii) 1-DOPA to dopamine; iv) dopamine to (S)-norcoclaurine; v) (S)-norcoclaurine to (S)/(R)-reticuline; vi) (R)-reticuline to salutaridine; vii) salutaridine to salutaridinol; viii) salutaridinol to salutaridinol-7-O-acetate; ix) salutaridinol-7-O-acetate to thebaine; x) thebaine to oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, and/or

buprenorphine.

(130) The method can further comprise selecting a cell comprising modulated levels of the desired polypeptide and growing such cell to obtain a plurality of cells.

(131) In some cases, the decrease expression of the thebaine synthesis polypeptide is desired. For example, an endogenous thebaine synthesis polypeptide may be silenced or knocked out. The polynucleotides encoding thebaine synthesis polypeptide may be used to produce a cell that has modified levels of expression of a desired polypeptide by gene silencing. For example, disclosed herein is a method of reducing the expression of thebaine synthesis polypeptide in a cell, comprising: (a) providing a cell expressing a desired polypeptide; and (b) silencing expression of the desired polypeptide in the cell. In some cases, the desired polypeptide can be a thebaine synthesis polypeptide.

(132) In some cases, a decreased expression of a purine permease is desired. For example, an endogenous purine permease may be silenced or knocked out. The polynucleotides encoding purine permease may be used to produce a cell that has modified levels of expression of a desired polypeptide by gene silencing. For example, disclosed herein is a method of reducing the expression of purine permease in a cell, comprising: (a) providing a cell expressing a desired polypeptide; and (b) silencing expression of the desired polypeptide in the cell. In some cases, the desired polypeptide can be a purine permease.

(133) Modifying the expression of endogenous genes may be achieved in a variety of ways. For example, antisense or RNA interference approaches may be used to down-regulate expression of the polynucleotides of the present disclosure, e.g., as a further mechanism for modulating cellular phenotype. That is, antisense sequences of the polynucleotides of the present disclosure, or subsequences thereof, may be used to block expression of naturally occurring homologous polynucleotide sequences. In particular, constructs comprising a desired polypeptide coding sequence, including fragments thereof, in antisense orientation, or combinations of sense and antisense orientation, may be used to decrease or effectively eliminate the expression of the desired polypeptide in a cell or plant and obtain an improvement in shelf life as is described herein. Accordingly, this may be used to “knock-out” the desired polypeptide or homologous sequences thereof. A variety of sense and antisense technologies, e.g., as set forth in Lichtenstein and Nellen (*Antisense Technology: A Practical Approach* IRL Press at Oxford University, Oxford, England, 1997), can be used. Sense or antisense polynucleotide can be introduced into a cell, where they are transcribed. Such polynucleotides can include both simple oligonucleotide sequences and catalytic sequences such as ribozymes.

(134) Other methods for a reducing or eliminating expression (i.e., a “knock-out” or “knockdown”) of a desired polypeptide (e.g., thebaine synthesis polypeptide and/or purine permease) in a transgenic cell or plant can be done by introduction of a construct which expresses an antisense of the desired polypeptide coding strand or fragment thereof. For antisense suppression, the desired polypeptide cDNA or fragment thereof is arranged in reverse orientation (with respect to the coding sequence) relative to the promoter sequence in the expression vector. Further, the introduced sequence need not always correspond to the full length cDNA or gene, and need not be identical to the cDNA or gene found in the cell or plant to be transformed.

(135) Additionally, the antisense sequence need only be capable of hybridizing to the target gene or RNA of interest. Thus, where the introduced polynucleotide sequence is of shorter length, a higher degree of homology to the endogenous transcription factor sequence will be needed for effective antisense suppression. While antisense sequences of various lengths can be utilized, in some embodiments, the introduced antisense polynucleotide sequence in the vector is at least 10, 20, 30, 40, 50, 100 or more nucleotides in length in certain embodiments. Transcription of an antisense construct as described results in the production of RNA molecules that comprise a sequence that is the reverse complement of the mRNA molecules transcribed from the endogenous gene to be repressed.

(136) Other methods for a reducing or eliminating expression can be done by introduction of a construct that expresses siRNA that targets a desired polypeptide (e.g., thebaine synthesis polypeptide and/or purine permease). In certain embodiments, siRNAs are short (20 to 24-bp) double-stranded RNA (dsRNA) with phosphorylated 5' ends and hydroxylated 3' ends with two overhanging nucleotides.

(137) Other methods for a reducing or eliminating expression can be done by insertion mutagenesis using the T-DNA of *Agrobacterium tumefaciens* or a selection marker cassette or any other non-sense DNA fragments. After generating the insertion mutants, the mutants can be screened to identify those containing the insertion in the thebaine synthesis polypeptide and/or purine permease (or other desired polypeptide) gene. Plants containing one or more transgene insertion events at the desired gene can be crossed to generate homozygous plant for the mutation, as described in Koncz et al., (Methods in Arabidopsis Research; World Scientific, 1992).

(138) Suppression of gene expression may also be achieved using a ribozyme. Ribozymes are RNA molecules that possess highly specific endoribonuclease activity. The production and use of ribozymes are disclosed in U.S. Pat. Nos. 4,987,071 and 5,543,508. Synthetic ribozyme sequences including antisense RNAs can be used to confer RNA cleaving activity on the antisense RNA, such that endogenous mRNA molecules that hybridize to the antisense RNA are cleaved, which in turn leads to an enhanced antisense inhibition of endogenous gene expression.

(139) A cell or plant gene may also be modified by using the Cre-lox system (for example, as described in U.S. Pat. No. 5,658,772). A cellular or plant genome can be modified to include first and second lox sites that are then contacted with a Cre recombinase. If the lox sites are in the same orientation, the intervening DNA sequence between the two sites is excised. If the lox sites are in the opposite orientation, the intervening sequence is inverted.

(140) In addition, silencing approach using short hairpin RNA (shRNA) system, and complementary mature CRISPR RNA (crRNA) by CRISPR/Cas system, and virus inducing gene silencing (VIGS) system may also be used to make down regulated or knockout of synthase mutants. Dominant negative approaches may also be used to make down regulated or knockout of desired polypeptides.

(141) The RNA-guided endonuclease can be derived from a clustered regularly interspersed short palindromic repeats (CRISPR)/CRISPR-associated (Cas) system. The CRISPR/Cas system can be a type I, a type II, or a type III system. Non-limiting examples of suitable CRISPR/Cas proteins include Cas3, Cas4, Cas5, Cas5e (or CasD), Cas6, Cas6e, Cas6f, Cas7, Cas8a1, Cas8a2, Cas8b, Cas8c, Cas9, Cas10, Cas10d, CasF, CasG, CasH, Csy1, Csy2, Csy3, Cse1 (or CasA), Cse2 (or CasB), Cse3 (or CasE), Cse4 (or CasC), Csc1, Csc2, Csa5, Csn2, Csm2, Csm3, Csm4, Csm5, Csm6, Cmr1, Cmr3, Cmr4, Cmr5, Cmr6, Csb1, Csb2, Csb3, Csx17, Csx14, Csx10, Csx16, CsaX, Csx3, Csz1, Csx15, Csf1, Csf2, Csf3, Csf4, and Cu1966.

(142) In general, CRISPR/Cas proteins comprise at least one RNA recognition and/or RNA binding domain. RNA recognition and/or RNA binding domains interact with guide RNAs. CRISPR/Cas proteins can also comprise nuclease domains (i.e., DNase or RNase domains), DNA binding domains, helicase domains, RNase domains, protein-protein interaction domains, dimerization domains, as well as other domains.

(143) The CRISPR/Cas-like protein can be a wild type CRISPR/Cas protein, a modified CRISPR/Cas protein, or a fragment of a wild type or modified CRISPR/Cas protein. The CRISPR/Cas-like protein can be modified to increase nucleic acid binding affinity and/or specificity, alter an enzyme activity, and/or change another property of the protein. For example, nuclease (i.e., DNase, RNase) domains of the CRISPR/Cas-like protein can be modified, deleted, or inactivated. Alternatively, the CRISPR/Cas-like protein can be truncated to remove domains that are not essential for the function of the fusion protein. The CRISPR/Cas-like protein can also be truncated or modified to optimize the activity of the effector domain of the fusion protein.

(144) One method to silence a desired gene (or a thebaine synthesis polypeptide and/or purine

permease gene) is virus induced gene silencing (known to the art as VIGS). In general, in plants infected with unmodified viruses, the viral genome is targeted. However, when viral vectors have been modified to carry inserts derived from host genes (e.g. portions of sequences encoding a desired polypeptide such as thebaine synthesis polypeptide and/or purine permease), the process is additionally targeted against the corresponding mRNAs. Thus disclosed is a method of producing a plant expressing reduced levels of a desired gene (such as thebaine synthesis polypeptide and/or purine permease) or other desired gene(s), the method comprising (a) providing a plant expressing a desired gene (e.g., a thebaine synthesis polypeptide and/or purine permease); and (b) reducing expression of the desired gene in the plant using virus induced gene silencing.

(145) Cell-Types

(146) The cells that can be used include but are not limited to plant or animal cells, fungus, yeast, algae, or bacterium. The cells can be prokaryotes or in some cases can be eukaryotes. For example, the cell can be a *Papaver somniferum* cell, *Saccharomyces cerevisiae*, *Yarrowia lipolytica*, or *Escherichia coli*, or any other cell disclosed throughout.

(147) In certain cases, the living cells are cells not naturally capable of producing thebaine (or other target morphinan alkaloids or morphinan alkaloid derivatives). In some cases, the cells are able to produce BIAs, such as thebaine, but at a low level. By implementation of the methods described herein, the cells can be modified such that the level of thebaine in the cells is higher relative to the level of thebaine produced in the unmodified cells. This can also allow the cells to produce higher levels of other target morphinan alkaloids or morphinan alkaloid derivatives.

(148) The levels of thebaine in the modified cells can be higher than the levels of thebaine in the unmodified cells.

(149) In some cases, the modified cell is capable of producing a substrate morphinan alkaloid, but not capable of naturally producing thebaine. The genetically modified cells in some cases are unable to produce a substrate morphinan alkaloid, and the substrate morphinan alkaloid is provided to the cells as part of the cell's growth medium. In this case, the genetically modified cell can process the substrate morphinan alkaloid into a desired product such as thebaine or other BIA.

(150) The genetically modified cell in some cases produces a substrate morphinan alkaloid when exogenously supplied with a base substrate, for example, supplied in a host cell growth medium. In some cases, the cell is capable of producing substrate morphinan alkaloid (or able to produce thebaine or other BIAs) only if a base substrate is included in host cell's growth medium.

(151) The genetically modified cell can comprise one or more enzyme capable of catalyzing one or more of the reactions: a sugar to L-tyrosine; L-tyrosine to L-DOPA; L-DOPA to Dopamine; Dopamine to (S)-Norcoclaurine; (S)-Norcoclaurine to (S)/(R)-Reticuline; (R)-Reticuline to Salutaridine; Salutaridine to Salutaridinol; or Salutaridinol to Salutaridinol-7-O-acetate.

(152) BIA Production Levels

(153) The genetic modifications to the cells described throughout can be made to produce BIAs (e.g., thebaine) over what would have been made without any genetic modifications. For example, compared to a non-genetically altered cell, the genetically modified cells described throughout can produce BIAs greater than 1.1 times (compared to the production levels of a non-genetically modified cell). In some cases, the genetically modified cells described throughout can produce BIAs greater than 1.2; 1.3; 1.4; 1.5; 1.6; 1.7; 1.8; 1.9; 2.0; 2.5; 3.0; 3.5; 4.0; 4.5; 5.0; 6.0; 7.0; 8.0; 9.0; 10.0; 12.5; 15.0; 17.5; 20.0; 25.0; 30.0; 50.0; 75.0; or 100.0 times compared to the production level of a non-genetically modified cell.

(154) In some cases, the BIA can be thebaine. In other cases, the BIA can be other BIAs described herein such as oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof.

(155) Methods of Making BIAs (or Morphinan Alkaloid, e.g., Thebaine)

(156) The genetically modified cells described throughout can be used to make BIAs, including

morphinan alkaloids, e.g., thebaine. A substrate morphinan alkaloid can be brought in contact with a thebaine synthesis polypeptide in a reaction mixture under reaction conditions permitting a thebaine synthesis polypeptide mediated reaction resulting in the conversion of the substrate morphinan alkaloid into thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives, under in vivo reaction conditions. Under such in vivo reaction conditions living cells are modified in such a manner that they produce morphinan alkaloids, e.g., thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives. Additionally, a purine permease can be present within the cells to increase the overall thebaine production titers.

(157) The BIAs (morphinan alkaloids, e.g., thebaine) produced may be recovered and isolated from the modified cells. The BIAs (morphinan alkaloids, e.g., thebaine) in some cases may be secreted into the media of a cell culture, in which BIA is extracted directly from the media. In some cases, the BIA may be within the cell itself, and the cells will need to be lysed in order to recover the BIA. In some instances, both cases may be true, where some BIAs are secreted and some remains within the cells. In this case, either method or both methods can be used.

(158) Accordingly, disclosed is a method for preparing a thebaine (or other BIAs), the method comprising: (a) providing a chimeric polynucleotide comprising as operably linked components: (i) a polynucleotide encoding a thebaine synthesis polypeptide; (ii) one or more polynucleotides capable of controlling expression in a host cell; (b) introducing the chimeric polynucleotide into a host cell that endogenously produces or is exogenously supplied with a substrate morphinan alkaloid; and (c) growing the host cell to produce the thebaine synthesis polypeptide and thebaine. The cell can also further comprise a purine permease. The method can further comprise recovering thebaine from the host cell.

(159) The substrate that can be used in these methods can be a substrate morphinan alkaloid. The substrate morphinan alkaloid can be salutaridine, salutaridinol, salutaridinol 7-O-acetate, or any combination thereof.

(160) The host cells can in some cases be capable of producing a substrate morphinan alkaloid, including salutaridine, salutaridinol, salutaridinol 7-O-acetate or any combination thereof. If the host cell cannot produce a substrate morphinan alkaloid, the substrate morphinan alkaloids, including salutaridine, salutaridinol and/or salutaridinol 7-O-acetate, are exogenously supplied to the host cells in order to produce a BIA.

(161) Disclosed herein is also a method of producing a BIA (or thebaine) comprising: (a) providing a host cell growth medium comprising a base substrate; (b) providing a host cell capable of (i) producing a thebaine synthesis polypeptide; and (ii) metabolizing the base substrate to form a substrate morphinan alkaloid; and (c) growing the host cell in the growth medium under conditions permitting a thebaine synthesis polypeptide mediated chemical reaction resulting in the conversion of the substrate morphinan alkaloid to form thebaine. Additionally, after (ii), the method can include producing a morphinan alkaloid derivative of thebaine by converting thebaine. In some instances, the method can further include isolating the morphinan alkaloid derivative of thebaine from the host cell culture. In some cases, the cells used in the method can comprise a purine permease.

(162) The thebaine synthesis polypeptides may further be used to produce thebaine or another target morphinan alkaloid or a morphinan alkaloid derivative of thebaine using processes involving growth of a host cell comprising the thebaine synthesis polypeptides in a growth medium comprising a base substrate. Therefore, disclosed herein is a method of using a thebaine synthesis polypeptide to produce thebaine, morphinan alkaloid, or morphinan alkaloid derivative of thebaine, in a cell grown in a growth medium comprising a base substrate, wherein the cell comprises the thebaine synthesis polypeptide. In some case, the cell can comprise a purine permease.

(163) The base substrate used in the methods to make a BIA can be a sugar. For example, the sugar can include but is not limited to glucose, fructose, galactose, mannose, or any combination thereof. The base substrate can also be a precursor compound of a substrate alkaloid morphinan including,

without limitation, glycerol, L-tyrosine, tyramine, L-3,4-dihydroxyphenyl alanine (L-DOPA), dopamine, or any combination thereof.

(164) The methods disclosed herein can make a morphinan alkaloid derivative of thebaine. For clarity these morphinan alkaloid derivative of thebaine are also BIAs and can include without limitation, oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof.

(165) The inclusion of thebaine synthesis polypeptides (and/or purine permease) in reaction mixtures comprising a morphinan alkaloid substrate can lead to substantially avoiding the synthesis of morphinan alkaloid by-products, which in the absence of a thebaine synthesis polypeptide (and/or purine permease) can accumulate at the expense of thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives. To the best of the inventors' knowledge, the current disclosure provides, for the first time, a methodology to use recombinant techniques to manufacture BIAs, thebaine or other target morphinan alkaloids or morphinan alkaloid derivatives via a process involving the use of thebaine synthesis polypeptides (and/or purine permease). The methodologies and compositions herein allow the synthesis of BIAs, thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives, at commercial levels, using cells that are not normally capable of synthesizing BIAs, thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives. Such cells may be used as a source of BIAs, thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives and may eventually be economically extracted from the cells itself or from culture media. The BIAs, thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives, produced can be useful inter alia in the manufacture of pharmaceutical compositions.

(166) Disclosed herein is also a method of making a morphinan alkaloid derivative of thebaine. This method can be performed by providing a substrate morphinan alkaloid and then contacting the substrate morphinan alkaloid with a thebaine synthesis polypeptide in a reaction mixture under reaction conditions that first permit a thebaine synthesis polypeptide mediated chemical reaction. This reaction results in the conversion of the substrate morphinan alkaloid to thebaine. If desired, then at least one additional chemical reaction can result in the conversion of thebaine to a morphinan alkaloid derivative of thebaine. The additional reaction can be mediated by a polypeptide.

(167) In order to convert the substrate alkaloid morphinan to thebaine, the substrate alkaloid morphinan is contacted with a thebaine synthesis polypeptide under reaction conditions permitting the conversion of the substrate alkaloid morphinan into thebaine. The thebaine synthesis polypeptide may be any thebaine synthesis polypeptide capable of mediating the conversion of the substrate alkaloid morphinan into thebaine. In some cases, more than one thebaine synthesis polypeptide may be used, e.g., one, two, three, four, five, six, seven, eight or more of the thebaine synthesis polypeptides or any functional variants or fragments thereof can be used. In some cases, a purine permease can be used in conjunction with the thebaine synthesis polypeptide to increase thebaine titers. In some cases, more than purine permease may be used, e.g., one, two, three, four, five, six, seven, eight or more of the purine permeases or any functional variants or fragments thereof can be used.

(168) The reaction conditions may be a reaction condition that permits the conversion of a substrate to a desired product. The reaction conditions can include in vivo or in vitro conditions, as hereinafter further detailed. The thebaine synthesis polypeptide, and other optional enzymes such as purine permease, SalAT, and SalR polypeptides, are provided in catalytic quantities. The reaction conditions can further include the presence of water and buffering agents. The reaction is can be conducted at neutral pH or mild basic or acidic pH (from approximately pH 5.5 to approximately pH 9.5). Further, other additives can be included in a reaction mixture, such as cofactors like NADPH and acetyl-CoA. In particular, reaction mixtures comprising SalAT further comprise Acetyl-CoA, and reaction mixtures comprising SalR further comprise NADPH.

(169) The substrate alkaloid morphinan can be converted to thebaine via one or more intermediate morphinan alkaloid compounds. In some cases, the substrate morphinan alkaloid can be salutaridinol and the intermediate morphinan alkaloid compound can be salutaridinol 7-O-acetate. In other cases, the substrate morphinan alkaloid can be salutaridine and the intermediate morphinan alkaloid compounds can be salutaridinol and salutaridinol 7-O-acetate.

(170) The time period during which a substrate morphinan alkaloid and intermediate alkaloid morphinan compound exist, once the reaction is started, can vary and can depend on the reaction conditions selected. Furthermore, an equilibrium between the substrate morphinan alkaloid, and thebaine, or morphinan alkaloid or morphinan alkaloid derivative, as well as the optional intermediate morphinan alkaloid compounds, can form, wherein the reaction can comprise various amounts of the substrate alkaloid compound and thebaine, or morphinan alkaloid or morphinan alkaloid derivative, and, optionally, one or more intermediate morphinan alkaloid compounds. The relative amounts of each of these compounds may vary depending on the reaction conditions selected. In general, in accordance herewith, the amount of thebaine, or morphinan alkaloid or morphinan alkaloid derivative, upon substantial completion of the reaction, present in the reaction mixture is at least 50% (w/w), at least 60% (w/w), at least 70% (w/w), at least 80% (w/w), at least 90% (w/w), at least 95% (w/w), or at least 99% (w/w) of the total morphinan alkaloid compounds in the reaction mixture.

(171) In general, the presence of thebaine synthesis polypeptide in the reaction mixture results in a more efficient conversion of the substrate morphinan alkaloid, i.e. when comparing conversion of a substrate morphinan alkaloid in a first reaction mixture comprising a thebaine synthesis polypeptide, with conversion in a second reaction mixture identical to the first reaction mixture, but for the presence of a thebaine synthesis polypeptide therein, conversion of the substrate morphinan alkaloid compound into thebaine occurs in a more efficient fashion in the first reaction mixture. Additionally, the presence of a purine permease in the reaction mixture results in a more efficient conversion of the substrate morphinan alkaloid into a BIA, such as thebaine. In this regard, a conversion is deemed “more efficient” if larger quantities of thebaine are obtained in the reaction mixture upon substantial completion of the reaction, and/or if thebaine accumulates in the reaction mixture at a faster rate. Thus, for example, in certain cases, in the presence of a thebaine synthesis polypeptide (and/or purine permease), few or no newly formed morphinan alkaloid reaction products, other than the target production such as thebaine, morphinan alkaloids, or morphinan alkaloid derivatives, accumulate in the reaction mixture. In other cases, few or no newly formed morphinan alkaloid compounds, other than the target products such as thebaine, morphinan alkaloids, or morphinan alkaloid derivatives and the intermediate morphinan alkaloid compounds salutaridinol and/or salutaridinol 7-O-acetate, accumulate in the reaction mixture.

(172) Referring now to FIG. 5, shown therein is, by way of example, that in some embodiments, in the absence (–) in a reaction mixture of thebaine synthesis polypeptide (TS), the morphinan alkaloid by-products m/z 330 (a) and m/z 330 (b) denoted in FIG. 5 can accumulate in such reaction mixture. Conversely, in the presence (+) of thebaine synthesis polypeptide (THS), few morphinan alkaloids other than the measured morphinan alkaloid thebaine, and in some embodiments salutaridinol and/or salutaridinol 7-O-acetate accumulate in the reaction mixture, notably no substantive quantities of the morphinan alkaloid b-products m/z 330 (a) and m/z 330 (b) denoted in FIG. 5 accumulate in the reaction mixture.

(173) In some instances, upon substantial completion of the reaction, the reaction mixture comprises thebaine, or morphinan alkaloid or morphinan alkaloid derivative, in a weight percentage of least 50% (w/w), at least 60% (w/w), at least 70% (w/w), at least 80% (w/w), at least 90% (w/w), at least 95% (w/w), or at least 99% (w/w) of total morphinan alkaloid compounds in the reaction mixture. In some other cases, upon substantial completion of the reaction, the reaction mixture comprises thebaine, or other target morphinan alkaloid or morphinan alkaloid derivative, salutaridinol and/or salutaridinol 7-O-acetate, together in a weight percentage of at least 60%

(w/w), at least 70% (w/w), at least 80% (w/w), at least 90% (w/w), at least 95% (w/w), or at least 99% (w/w) of total morphinan alkaloid compounds in the reaction mixture. In some cases, upon substantial completion of the reaction, the reaction mixture comprises morphinan alkaloid compounds other than thebaine (or other target morphinan alkaloid or morphinan alkaloid derivative), in a weight percentage of less than 25% (w/w), less than 20% (w/w), less than 15% (w/w), less than 10% (w/w), less than 5% (w/w), or less than 1% (w/w) of total morphinan alkaloids in the reaction mixture. In other cases, upon substantial completion of the reaction, the reaction mixture comprises morphinan alkaloid compounds other than thebaine, or other target morphinan alkaloid or morphinan alkaloid derivative, salutaridinol and/or salutaridinol acetate in a weight percentage of less than 15% (w/w), less than 10% (w/w), less than 5% (w/w), or less than 1% (w/w) of total morphinan alkaloids in the reaction mixture. In some cases, upon substantial completion of the reaction, the reaction mixture comprises morphinan alkaloid compounds m/z 330 (a) and/or m/z 330 (b) together in a weight percentage of less than 25% (w/w), less than 20% (w/w), less than 15% (w/w), less than 10% (w/w), less than 5% (w/w), or less than 1% (w/w) of total morphinan alkaloids in the reaction mixture.

(174) The thebaine synthesis polypeptide (and/or purine permease) can mediate the conversion by preventing the formation of reaction products, other than thebaine, or in some embodiments, reaction products other than salutaridinol or salutaridinol 7-O-acetate, in the reaction mixture, and in particular by preventing the formation of morphinan alkaloid compounds m/z 330 (a) and/or m/z 330 (b). Furthermore, the thebaine synthesis polypeptide (and/or purine permease) may act by preventing certain oxidation reactions, for example, oxidation by molecular oxygen (O₂) and/or oxidation by water (H₂O). In particular, the thebaine synthesis polypeptide (and/or purine permease) may act by preventing the oxidation of carbon atom C₅ to form alkaloid morphinan compound m/z 330 (b), and/or by preventing the oxidation of carbon atom C₁₄ to form alkaloid morphinan compound m/z 330 (a), at the expense of the formation of thebaine.

(175) The thebaine, other target morphinan alkaloid or morphinan alkaloid derivative formed from the methods described herein can be isolated from the reaction mixture. Methods for the isolation of morphinan compounds can be selected as desired. For example, the thebaine (or other target morphinan alkaloid or morphinan alkaloid derivative) may be isolated from aqueous mixtures using a liquid/liquid extraction with toluene (see: United States Patent Application No 2002/0106761) or ethyl acetate or ethyl acetate (see: Lenz, R., and Zenk M. H., 1995, *Eur. J. Biochem*, 233, 132-139). Further techniques and guidance for the isolation of thebaine from reaction mixture may be found in, for example, Hansen, S. H. *J. Sep. Sci.* (2009), 32 (5-6), 825-834; U.S. Pat. No. 7,094,351; and in (Rinner, U., and Hudlicky, J., 2012 *Top. Cur. Chem.* 209: 33-66).

(176) BIA Production Levels

(177) The methods described throughout can be used to produce BIAs over what would have been made without any genetic modifications. For example, the methods described throughout can produce BIAs greater than 1.1 times (compared to levels of standard methods, e.g., methods that do not use genetically modified cells). In some cases, the genetically modified cells described throughout can produce BIAs greater than 1.2; 1.3; 1.4; 1.5; 1.6; 1.7; 1.8; 1.9; 2.0; 2.5; 3.0; 3.5; 4.0; 4.5; 5.0; 6.0; 7.0; 8.0; 9.0; 10.0; 12.5; 15.0; 17.5; 20.0; 25.0; 30.0; 50.0; 75.0; or 100.0 times compared to standard methods (including methods that do not use genetically modified cells).

(178) In some cases, the BIA can be thebaine. In other cases, the BIA can be other BIAs described herein such as oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof.

(179) Other Methods and Uses

(180) Mutagenesis

(181) Mutagenesis can be used to create cells and plants that comprise improved polypeptides. The

improved polypeptides can be used to increase the production of BIAs (or certain components within the BIA pathway) from genetically modified cells and plants.

(182) For example, cells, such as plant seed cells can be used to perform the mutagenesis. Mutagenic agents that can be used are chemical agents, including without limitation, base analogues, deaminating agents, alkylating agents, intercalating agents, transposons, bromine, sodium azide, ethyl methanesulfonate (EMS) as well as physical agents, including, without limitation, radiation, such as ionizing radiation and UV radiation. After mutagenesis of a cell, the activity of the desired polypeptide (e.g., a thebaine synthesis polypeptide and/or purine permease) can be measured and compared to a cell that was not mutagenized.

(183) For example, the following is a method to alter the activity of a thebaine synthesis polypeptide by mutagenesis of a seed setting plant. Disclosed herein is a method for producing a seed setting plant comprising modulated levels of expression of thebaine synthesis polypeptide and/or purine permease in a plant cell naturally expressing thebaine synthesis polypeptide and/or purine permease, the method comprising: (a) providing a seed setting plant naturally expressing thebaine synthesis polypeptide and/or purine permease; (b) mutagenizing seed of the plant to obtain mutagenized seed; (c) growing the mutagenized seed into the next generation mutagenized plants capable of setting the next generation seed; and (d) obtaining the next generation seed, or another portion of the mutagenized plants, and analyzing if the next generation plants or next generation seed exhibits modulated thebaine synthesis polypeptide and/or purine permease expression.

(184) The analysis of the mutagenized cells or plants can be done in several ways. After a plurality of generations of plants and/or seeds is obtained, the portions of plants and/or seeds obtained in any or all of such generations can be analyzed. The analysis can be performed by comparison of expression levels of, for example, RNA levels or protein, in non-mutagenized (wild type) plants or seed, with expression levels in mutagenized plants or seed. The analysis can be performed by analysis of heteroduplex formation between wildtype DNA and mutated DNA. In some instances, the analysis can comprise: (i) extracting DNA from mutated plants; (ii) amplifying a portion of the DNA comprising a polynucleotide encoding thebaine synthesis polypeptide and/or purine permease to obtain amplified mutated DNA; (iii) extracting DNA from wild type plants; (iv) mixing the DNA from wild type plants with the amplified mutated DNA and form a heteroduplexed polynucleotide; (v) incubating the heteroduplexed polynucleotide with a single stranded restriction nuclease capable of restricting at a region of the heteroduplexed polynucleotide that is mismatched; and (vi) determining the site of mismatch in the heteroduplexed polynucleotide.

(185) The methods that modify the expression levels of thebaine synthesis polypeptide and/or purine permease (or other desired gene) may result in modulations in the levels of plant morphinan alkaloids, in plants including but not limited to, thebaine, oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof.

(186) Genotyping

(187) The polynucleotides encoding thebaine synthesis polypeptide and/or purine permease can be used to genotype plants. In general, genotyping provides a means of distinguishing homologs of a chromosome pair and can be used to identify segregants in subsequent generations of a plant population. Molecular marker methodologies can be used for phylogenetic studies, characterizing genetic relationships among plant varieties, identifying crosses or somatic hybrids, localizing chromosomal segments affecting monogenic traits, map based cloning, and the study of quantitative inheritance. See, e.g., *Plant Molecular Biology: A Laboratory Manual*, Chapter 7, Clark, Ed., Springer-Verlag, Berlin (1997). For molecular marker methodologies, see generally, *The DNA Revolution* by Andrew H. Paterson 1996 (Chapter 2) in: *Genome Mapping in Plants* (ed. Andrew H. Paterson) by Academic Press/R. G. Landis Company, Austin, Tex., pp. 7-21. The particular method of genotyping in accordance with the disclosure may involve the employment of any molecular marker analytic technique including, but not limited to, restriction fragment length

polymorphisms (RFLPs). RFLPs reflect allelic differences between DNA restriction fragments caused by nucleotide sequence variability. As is known to those of skill in the art, RFLPs are typically detected by extraction of plant genomic DNA and digestion of the genomic DNA with one or more restriction enzymes. Typically, the resulting fragments are separated according to size and hybridized with a nucleic acid probe. Restriction fragments from homologous chromosomes are revealed. Differences in fragment size among alleles represent an RFLP. Thus, disclosed is a means to follow segregation of a portion of genomic DNA encoding thebaine synthesis polypeptide, and/or purine permease as well as chromosomal polynucleotides genetically linked to these thebaine synthesis polypeptides and/or purine permease encoding polynucleotides using such techniques as RFLP analysis. Linked chromosomal nucleic sequences are within 50 centiMorgans (cM), often within 40 or 30 cM, preferably within 20 or 10 cM, more preferably within 5, 3, 2, or 1 cM of a genomic polynucleotide encoding thebaine synthesis polypeptide and/or purine permease. Thus, the thebaine synthesis polypeptide and/or purine permease encoded by polynucleotide sequences may be used as markers to evaluate in a plant population the segregation of polynucleotides genetically linked thereto.

(188) Some plants that can be useful to genotype with the method disclosed herein are such a plant preferably belongs to the plant genus *Papaver* or the plant species *Papaver bracteatum*, *Papaver somniferum*, *Papaver cylindricum*, *Papaver decaisnei*, *Papaver fugax*, *Papaver nudicale*, *Papaver oreophyllum*, *Papaver orientate*, *Papaver paeonifolium*, *Papaver persicum*, *Papaver pseudo-orientale*, *Papaver rhoeas*, *Papaver rhopalotheca*, *Papaver armeniacum*, *Papaver setigerum*, *Papaver tauricolum*, and *Papaver triniaefolium*.

(189) Polynucleotide probes employed for molecular marker mapping of plant nuclear genomes can selectively hybridize, under selective hybridization conditions, to a genomic sequence encoding a thebaine synthesis polypeptide and/or purine permease (or other gene of interest). The probes can be selected from the polynucleotides encoding thebaine synthesis polypeptides and/or purine permease disclosed herein. In other words, the probes can be designed to be complementary to at least a portion of the polynucleotide sequences disclosed herein, e.g., the polynucleotide sequences for the thebaine synthesis polypeptide (e.g., SEQ ID NO. 19) and/or purine permease (e.g., any one of SEQ ID NOs. 36, 38, 39, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, or 64). In some cases, SEQ ID NOs. 6, 31, 32, 35, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, and/or 63 can be particularly useful for designing a probe.

(190) The probes can be cDNA probes. The probes can also be at least 5, 6, 7, 8 or 9 bases in length, more preferably at least 10, 15, 20, 25, 30, 35, 40, 50 or 100 bases in length. Generally, however, the probes are less than about 1 kilobase in length. For example, the probes can be anywhere from 10 bp to 30 bps in length. The probes can be single copy probes that hybridize to a unique locus in a haploid plant chromosome complement. In some cases, the polynucleotide is a primer. Some exemplary restriction enzymes employed in RFLP mapping are EcoRI, EcoRV, and SstI. As used herein the term "restriction enzyme" includes reference to a composition that recognizes and, alone or in conjunction with another composition, cleaves a polynucleotide at a specific polynucleotide sequence.

(191) Other methods of differentiating polymorphic (allelic) variants of the polynucleotides can be used by utilizing molecular marker techniques, including, without limitation: 1) single stranded conformation analysis (SSCP); 2) denaturing gradient gel electrophoresis (DGGE); 3) RNase protection assays; 4) allele-specific oligonucleotides (ASOs); 5) the use of proteins which recognize nucleotide mismatches, such as the *E. coli* mutS protein; and 6) allele-specific PCR. Other approaches based on the detection of mismatches between the two complementary DNA strands include, without limitation, clamped denaturing gel electrophoresis (CDGE); heteroduplex analysis (HA), and chemical mismatch cleavage (CMC). Thus, disclosed is a method of genotyping comprising contacting, under stringent hybridization conditions, a sample suspected of comprising a nucleic acid encoding a thebaine synthesis polypeptide and/or purine permease (or other

interested genes), with a nucleic acid probe capable of hybridizing thereto. For example, a sample can comprise plant material, including, a sample suspected of comprising a *Papaver somniferum* polynucleotide encoding a thebaine synthesis polypeptide and/or purine permease (e.g., gene, mRNA). The polynucleotides probe selectively hybridizes, under stringent conditions, to a subsequence of the polynucleotide encoding thebaine synthesis polypeptide and/or purine permease comprising a polymorphic marker. Selective hybridization of the polynucleotide probe to the polymorphic marker polynucleotide yields a hybridization complex. Detection of the hybridization complex indicates the presence of that polymorphic marker in the sample. In preferred embodiments, the polynucleotide probe comprises a portion of a polynucleotide encoding thebaine synthesis polypeptide and/or purine permease.

(192) In Vitro Synthesis of BIAs

(193) The BIAs disclosed herein can be synthesized outside of a cell. For example, one or more of the enzymes (e.g., including the thebaine synthesis polypeptide and/or purine permease) disclosed throughout can be isolated and put into the media outside of a cell. These enzymes can then catalyze the reaction from a substrate into a product outside of a cell. In the case of the thebaine synthesis polypeptide, the substrate can sometimes be salutaridinol 7-O-acetate and the product can be thebaine.

(194) Enzymes and Other Components

(195) For in vitro synthesis of a BIA (such as thebaine), a method can include contacting in vitro a substrate with one or more enzymes that are able to convert the substrate. The one or more enzymes can be any of the following: a tyrosine hydroxylase (TYR); DOPA decarboxylase (DODC); norcoclaurine synthase (NCS); 6-O-Methyltransferase (6OMT); coclaurine N-methyltransferase (CNMT), cytochrome P450 N-methylcoclaurine hydroxylase (NMCH), and 4-O-methyltransferase (4OMT); cytochrome P450 reductase (CPR), salutaridine synthase (SAS); salutaridine reductase (SalR); salutaridinol-7-O-acetyltransferase (SalAT); purine permease (PUP); thebaine synthesis polypeptide (THS); a codeine O-demethylase (CODM); a thebaine 6-O-demethylase (T6ODM); a codeinone reductase (COR); or any combination thereof.

(196) Pure forms of the enzymes necessary for this method may be obtained by isolation of the polypeptide from certain sources including, without limitation, the plants. For example, the plants that can be used include plants from the genus *Papaver*, or from the plant species such as *Papaver somniferum*. Other plants can include without limitation, plant species belonging to the plant families of Eupteleaceae, Lardizabalaceae, Circaeasteraceae, Menispermaceae, Berberidaceae, Ranunculaceae, and Papaveraceae (including those belonging to the subfamilies of Pteridophylloideae, Papaveroideae and Fumarioideae) and further includes plants belonging to the genus *Argemone*, including *Argemone mexicana* (Mexican Prickly Poppy), plants belonging to the genus *Berberis*, including *Berberis thunbergii* (Japanese Barberry), plants belonging to the genus *Chelidonium*, including *Chelidonium majus* (Greater Celandine), plants belonging to the genus *Cissampelos*, including *Cissampelos mucronata* (Abuta), plants belonging to the genus *Cocculus*, including *Cocculus trilobus* (Korean Moonseed), plants belonging to the genus *Corydalis*, including *Corydalis chelanthifolia* (Ferny Fumewort), *Corydalis cava*; *Corydalis ochotenis*; *Corydalis ophiocarpa*; *Corydalis platycarpa*; *Corydalis tuberosa*; and *Cordyalis bulbosa*, plants belonging to the genus *Eschscholzia*, including *Eschscholzia californica* (California Poppy), plants belonging to the genus *Glaucium*, including *Glaucium flavum* (Yellowhorn Poppy), plants belonging to the genus *Hydrastis*, including *Hydrastis canadensis* (Goldenseal), plants belonging to the genus *Jeffersonia*, including *Jeffersonia diphylla* (Rheumatism Root), plants belonging to the genus *Mahonia*, including *Mahonia aquifolium* (Oregon Grape), plants belonging to the genus *Menispermum*, including *Menispermum canadense* (Canadian Moonseed), plants belonging to the genus *Nandina*, including *Nandina domestica* (Sacred Bamboo), plants belonging to the genus *Nigella*, including *Nigella sativa* (Black Cumin), plants belonging to the genus *Papaver*, including *Papaver bracteatum* (Persian Poppy), *Papaver somniferum*, *Papaver cylindricum*, *Papaver*

decaisnei, *Papaver fugax*, *Papaver nudicale*, *Papaver oreophyllum*, *Papaver orientate*, *Papaver paeonifolium*, *Papaver persicum*, *Papaver pseudo-orientale*, *Papaver rhoeas*, *Papaver rhopalotheca*, *Papaver armeniacum*, *Papaver setigerum*, *Papaver tauricolum*, and *Papaver triniaefolium*, plants belonging to the genus *Sanguinaria*, including *Sanguinaria canadensis* (Bloodroot), plants belonging to the genus *Stylophorum*, including *Stylophorum diphyllum* (Celandine Poppy), plants belonging to the genus *Thalictrum*, including *Thalictrum flavum* (Meadow Rue), plants belonging to the genus *Tinospora*, including *Tinospora cordifolia* (Heartleaf Moonseed), plants belonging to the genus *Xanthoriza*, including *Xanthoriza simplicissima* (Yellowroot) and plants belonging to the genus *Romeria* including *Romeria carica*. The enzymes of the methods can also be prepared with recombinant techniques using a polynucleotide encoding the enzymes. For example, a thebaine synthesis polypeptide can be made recombinantly by expressing a sequence that is substantially identical to SEQ ID NO. 19, derivatives, fragments, or variants thereof. In a preferred embodiment, SEQ ID NO. 19, derivatives, fragments, or variants can be expressed to make a thebaine synthesis polypeptide. The expressed thebaine synthesis polypeptides can be recovered and used to make BIAs in vitro. Disclosed herein is a method for preparing a thebaine synthesis polypeptide, the method comprising: (a) introducing into a host cell a heterologous polynucleotide comprising operably linked (i) polynucleotides encoding a thebaine synthesis polypeptide; and (ii) one or more a polynucleotides capable of controlling expression in the host cell; (b) growing the host cell to produce the thebaine synthesis polypeptide; and (c) recovering the thebaine synthesis polypeptide from the host cell. A polynucleotide encoding for a thebaine synthesis polypeptide may be obtained in accordance herewith include, without limitation, from plant species listed above.

(197) Additionally, a purine permease can be made recombinantly by expressing a sequence that is substantially identical to any one of SEQ ID NOs. 35 to 64, derivatives, fragments, or variants thereof. In a preferred embodiment, a sequence that is substantially identical to any one of SEQ ID NOs. 35 to 64, derivatives, fragments, or variants can be expressed to make a purine permease. The expressed purine permease can be recovered and used to make BIAs in vitro. Disclosed herein is a method for preparing a purine permease, the method comprising: (a) introducing into a host cell a heterologous polynucleotide comprising operably linked (i) polynucleotides encoding a purine permease; and (ii) one or more a polynucleotides capable of controlling expression in the host cell; (b) growing the host cell to produce the purine permease; and (c) recovering the purine permease from the host cell. A polynucleotide encoding for a purine permease may be obtained in accordance herewith include, without limitation, from plant species listed above.

(198) The polynucleotide sequences described herein (e.g., SEQ ID NO. 19 or any other THS sequence disclosed throughout) encoding thebaine synthesis polypeptides (or any other desired genes) can be used to screen the genomes of plant species and other organisms, for the presence of polynucleotides encoding thebaine synthesis polypeptides (or other desired genes). For example, SEQ ID NO. 19 can be used to screen for other thebaine synthesis polypeptides. Upon identifying polynucleotides encoding thebaine synthesis polypeptides (or other desired genes), the genes can be isolated and used to prepare additional thebaine synthesis polypeptides (or other desired polypeptides). Sequences that are substantially identical can also be used.

(199) The polynucleotide sequences described herein (e.g., any one of SEQ ID NOs. 36, 38, 39, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, and/or 64) encoding a purine permease (or any other desired genes) can be used to screen the genomes of plant species and other organisms, for the presence of polynucleotides encoding a purine permease (or other desired genes). For example, SEQ ID NO. 36 can be used to screen for other purine permeases. Additionally, any one of SEQ ID NOs. 38, 39, 42, 44, 46, 48, 50, 52, 54, 56, 58, 60, 62, and/or 64 can be used to screen for other purine permeases. Upon identifying polynucleotides encoding purine permeases (or other desired genes), the genes can be isolated and used to prepare additional purine permeases (or other desired polypeptides). Sequences that are substantially identical can also be used.

(200) The preparation (used in the in vitro method) can also comprise a mixture of different thebaine synthesis polypeptides and/or purine permeases (e.g., from different sources or from a single source that expresses multiple thebaine synthesis polypeptides and/or purine permeases). The thebaine synthesis polypeptide and/or purine permease is polypeptide obtainable or obtained from the plant genus *Papaver*, or plant species *Papaver somniferum*. The thebaine synthesis polypeptide can be a polypeptide set forth in any one of SEQ ID NO. 6 or 31. In some cases, SEQ ID NO. 6 or 31 is a thebaine synthesis polypeptide. In some cases, SEQ ID NO. 32, 80, or 81 is a thebaine synthesis polypeptide. The purine permease can be a polypeptide set forth in SEQ ID NO. 35. In some cases, SEQ ID NO. 35 is a purine permease. In other cases, the purine permease can be any purine permease described throughout, such as those encoded by any one of SEQ ID NOs. 37, 41, 43, 45, 47, 49, 51, 53, 55, 57, 59, 61, and/or 63. Substantially identical polypeptides, derivatives, functional fragments, and variants can be used. The preparation can also be substantially pure of other proteins that are not thebaine synthesis polypeptides and/or purine permeases.

(201) Polypeptides that can assist the thebaine synthesis polypeptides can also be used. Any of the polypeptides that assist the thebaine synthesis polypeptide described throughout can be used. For example, the in vitro method can comprise one or more polypeptides that are encoded by a polynucleotide that is substantially identical to SEQ ID NO. 18; SEQ ID NO. 20; SEQ ID NO. 21; SEQ ID NO. 22; SEQ ID NO. 23; SEQ ID NO. 24; SEQ ID NO. 25; SEQ ID NO. 26; SEQ ID NO. 27; SEQ ID NO. 28; SEQ ID NO. 29; SEQ ID NO. 30; SEQ ID NO. 36; SEQ ID NO. 38; SEQ ID NO. 42; SEQ ID NO. 44; SEQ ID NO. 46; SEQ ID NO. 48; SEQ ID NO. 50; SEQ ID NO. 52; SEQ ID NO. 54; SEQ ID NO. 56; SEQ ID NO. 58; SEQ ID NO. 60; SEQ ID NO. 62; or SEQ ID NO. 64. In some instances, the in vitro method can comprise one or more polypeptides that are substantially identical to SEQ ID NO. 5; SEQ ID NO. 7; SEQ ID NO. 8; SEQ ID NO. 9; SEQ ID NO. 10; SEQ ID NO. 11; SEQ ID NO. 12; SEQ ID NO. 13; SEQ ID NO. 14; SEQ ID NO. 15; SEQ ID NO. 16; SEQ ID NO. 17; SEQ ID NO. 35; SEQ ID NO. 37; SEQ ID NO. 41; SEQ ID NO. 43; SEQ ID NO. 45; SEQ ID NO. 47; SEQ ID NO. 49; SEQ ID NO. 51; SEQ ID NO. 53; SEQ ID NO. 55; SEQ ID NO. 57; SEQ ID NO. 59; SEQ ID NO. 61; or SEQ ID NO. 63.

(202) Also disclosed herein is an isolated polynucleotide comprising a polynucleotide sequence encoding a thebaine synthesis polypeptide and/or purine permease. In some cases, the polynucleotides encoding a thebaine synthesis polypeptide and/or purine permease can be obtainable or obtained from the plant genus *Papaver* or from the plant species *Papaver somniferum*.

(203) The polynucleotide encoding a thebaine synthesis polypeptide and/or purine permease (or other desired enzyme) can be isolated or prepared using any desirable methodology. For example, a polynucleotide encoding a thebaine synthesis polypeptide and/or purine permease can be obtained or prepared following identification and isolation of a thebaine synthesis polypeptide, as further described in detail in Example 1.

(204) The thebaine synthesis polypeptide and/or purine permease (or other desired enzyme) may be recovered, isolated and separated from other host cell components by a variety of different protein purification techniques including, e.g. ion-exchange chromatography, size exclusion chromatography, affinity chromatography, hydrophobic interaction chromatography, reverse phase chromatography, gel filtration, etc. Further general guidance with respect to protein purification may for example be found in: Cutler, P. Protein Purification Protocols, Humana Press, 2004, Second Ed.

(205) The thebaine synthesis polypeptide may be any thebaine synthesis polypeptide capable of mediating the conversion of the substrate alkaloid morphinan into thebaine. In some cases, more than one thebaine synthesis polypeptide may be used, e.g., one, two, three, four, five, six, seven, eight or more of the thebaine synthesis polypeptides or any functional variants or fragments thereof can be used.

(206) The purine permease may be any purine permease capable of mediating increasing thebaine titers in a cell. In some cases, more than one purine permeases may be used, e.g., one, two, three, four, five, six, seven, eight or more of the purine permeases or any functional variants or fragments thereof can be used.

(207) Disclosed herein is a preparation comprising one or more isolated thebaine synthesis polypeptides and/or purine permeases. Such preparations generally are aqueous preparations containing additionally, for example, buffering agents, salts, reducing agents, and stabilizing elements to maintain solubility and stability of the thebaine synthesis polypeptides and/or purine permeases. The preparation can be substantially free of proteins other than thebaine synthesis polypeptides and/or purine permeases. Thus, for example, in some preparations the thebaine synthesis polypeptide comprises at least 90% (w/w), at least 95% (w/w), at least 96% (w/w), at least 97% (w/w), at least 98% (w/w), or at least 99% (w/w) of the protein constituents in a preparation. The preparation can also comprise a single thebaine synthesis polypeptide and is substantially free of other thebaine synthesis polypeptides. Thus, for example, in some preparations a single thebaine synthesis polypeptide comprises at least 90% (w/w), at least 95% (w/w), at least 96% (w/w), at least 97% (w/w), at least 98% (w/w), or at least 99% (w/w) of the thebaine synthesis polypeptide constituents in the preparation.

(208) In some preparations the purine permease comprises at least 90% (w/w), at least 95% (w/w), at least 96% (w/w), at least 97% (w/w), at least 98% (w/w), or at least 99% (w/w) of the protein constituents in a preparation. The preparation can also comprise a single purine permease and is substantially free of other purine permeases. Thus, for example, in some preparations a single purine permease comprises at least 90% (w/w), at least 95% (w/w), at least 96% (w/w), at least 97% (w/w), at least 98% (w/w), or at least 99% (w/w) of the purine permeases constituents in the preparation.

(209) Methods of Making BIA Outside of a Cell

(210) For in vitro synthesis of a BIA (such as thebaine), a method can include (a) contacting a substrate that is capable of being converted by one or more enzymes, where the one or more enzymes comprise a tyrosine hydroxylase (TYR); DOPA decarboxylase (DODC); norcoclaurine synthase (NCS); 6-O-Methyltransferase (6OMT); coclaurine N-methyltransferase (CNMT), cytochrome P450 N-methylcoclaurine hydroxylase (NMCH), and 4-O-methyltransferase (4OMT); cytochrome P450 reductase (CPR), salutaridine synthase (SAS); salutaridine reductase (SalR); salutaridinol-7-O-acetyltransferase (SalAT); purine permease (PUP); or any combination thereof and (b) contacting the product of (a) with one or more of a thebaine synthesis polypeptide; a codeine O-demethylase (CODM); a thebaine 6-O-demethylase (T6ODM); a codeinone reductase (COR); or any combination thereof.

(211) One or more the above mentioned enzymes may also be within a living cell. For example, TYR may be outside the cell while the rest of the enzymes may be contained with the cell. In another example, TYR, DODC, NCS, 6OMT, CNMT, NMCH, 4OMT; CPR, SAS, SalR, PUP, and SalAT are contained within a living cell, while a thebaine synthesis polypeptide (THS) is outside the cell in the medium. Further, CODM, T6ODM, and COR can also be outside the cell in the medium.

(212) The substrate can be any of the substrates disclosed throughout, including but not limited to a sugar (e.g., glucose), glycerol, L-tyrosine, tyramine, L-3,4-dihydroxyphenyl alanine (L-DOPA), alcohol, dopamine, or any combination thereof.

(213) The substrate can be contacted with one enzyme before it is contacted with another, in a sequential manner. For example, the substrate can be contacted with salutaridine synthase (SAS) before it is contacted with a thebaine synthesis polypeptide.

(214) After the BIA is formed, the method can comprise recovering the BIA.

(215) In the case of making thebaine, some of the methods can include bringing a substrate morphinan alkaloid in contact with a thebaine synthesis polypeptide in a reaction mixture under

reaction conditions permitting a thebaine synthesis polypeptide mediated reaction resulting in the conversion of the substrate morphinan alkaloid into thebaine under in vitro reaction conditions. Under such in vitro reaction conditions the initial reaction constituents are provided in a more or less isolated form, and the constituents are mixed under conditions that permit the reaction to substantially proceed.

(216) One or more plant cells or microorganisms can also be co-cultured with one or more different substrates to provide necessary reaction constituents. If one or more plant cells or microorganisms are used in these in vivo reactions, the necessary reaction constituents can be secreted.

(217) Should substrate morphinan alkaloids be used, substrate morphinan alkaloids may be purchased as a substantially pure chemical compound, chemically synthesized from precursor compounds, or isolated from certain sources including from the plants, such as from the plant genus *Papaver*, or from the plant species such as *Papaver somniferum*. Such compounds may be compounds provided with a high degree of purity, for example, a purity of at least 75%, 80%, 85%, 90%, 95%, or 99% or more. Other plant species that may be used to obtain the substrate morphinan alkaloid include, without limitation, plant species belonging to the plant families described above.

(218) With regards to making thebaine, the thebaine synthesis polypeptide may be used to produce thebaine (or other target morphinan alkaloids or morphinan alkaloid derivatives). Accordingly, described is a use of a thebaine synthesis polypeptide to make thebaine (or other target morphinan alkaloids or morphinan alkaloid derivatives). Additionally, a purine permease can also be used alone or in combination with a thebaine synthesis polypeptide to make substantial amounts of thebaine.

(219) The thebaine synthesis polypeptide and/or purine permease can be included in a reaction mixture in which a substrate alkaloid morphinan is converted to thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives. In accordance herewith in order to perform a reaction under in vitro conditions, a substrate morphinan alkaloid is brought in contact with catalytic quantities of thebaine synthesis polypeptide, under reaction conditions permitting a chemical conversion of the substrate morphinan alkaloid into thebaine, or other target morphinan alkaloids or morphinan alkaloid derivatives. In some cases, the agents are brought in contact with each other and mixed to form a mixture. The mixture can be an aqueous mixture comprising water and further optionally additional agents to mediate the reaction, including buffering agents, salts, pH-modifying agents and co-factors such as acetyl-CoA and NADPH. The reaction may be performed at a range of different temperatures, including between about 18° C. and about 50° C., e.g., between about 18° C. and about 45° C., about 25° C. to about 40° C., about 32° C. to about 38° C., or at about 37° C. The mixture can also comprise one or more purine permeases.

(220) The reaction mixture can further comprise SalAT and/or SalR. Polynucleotide sequences of SalAT and SalR that can be used are those identified as SEQ ID NO. 1 and SEQ ID NO. 3, respectively, or sequences that are substantially identical. These sequences may be used to in recombinant techniques to prepare SalAT and/or SalR polypeptides and polynucleotides encoding SalAT and/or SalR. Alternatively SalAT and/or SalR may be obtained from certain sources, including from a plant, including the plants disclosed herein.

(221) Upon substantial completion of the in vitro reaction a reaction mixture comprising thebaine is obtained. In some cases, other BIAs can be obtained. Thebaine or other BIA can then be isolated from the reaction mixture. In some cases, thebaine is left in the reaction mixture to form other target morphinan alkaloids or morphinan alkaloid derivatives (or other BIAs). In some cases, in order to form other target morphinan alkaloids or morphinan alkaloid derivatives, additional enzymes must be added to the reaction mixture.

(222) BIA Production Levels

(223) The in vitro methods described throughout can be made to produce BIAs over what would have been made with standard methods (including methods that do not use a thebaine synthesis polypeptide and/or purine permease and/or other polypeptides described throughout). For example,

the in vitro methods described throughout can produce BIAs greater than 1.1 times (compared to standard methods, including methods that do not use a thebaine synthesis polypeptide and/or purine permease and/or other polypeptides described throughout). In some cases, the in vitro methods described throughout can produce BIAs greater than 1.2; 1.3; 1.4; 1.5; 1.6; 1.7; 1.8; 1.9; 2.0; 2.5; 3.0; 3.5; 4.0; 4.5; 5.0; 6.0; 7.0; 8.0; 9.0; 10.0; 12.5; 15.0; 17.5; 20.0; 25.0; 30.0; 50.0; 75.0; or 100.0 times compared to the production level of standard methods (including methods that do not use a thebaine synthesis polypeptide and/or purine permease and/or other polypeptides described throughout).

(224) In some cases, the BIA can be thebaine. In other cases, the BIA can be other BIAs described herein such as oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof.

(225) Exemplary Uses of the BIAS

(226) Preparations of BIAs (e.g., thebaine) obtained may be used for a variety of compositions and methods. The BIAs can be isolated and sold as purified products. Or these purified products can be a feedstock to make additional BIAs or morphinan alkaloids.

(227) Derivative morphinan alkaloid compounds may be used to manufacture medicinal compounds. Thus, for example when considering thebaine, it be converted to a derivative morphinan alkaloid compound selected from oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone, naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof.

(228) Accordingly, in one aspect, disclosed is a use of thebaine (or other BIA) as a feedstock compound in the manufacture of a medicinal compound.

(229) The medicinal compound can be a natural derivative morphinan alkaloid compound or, in some cases, a semi-synthetic derivative morphinan alkaloid compound. For example, thebaine may be converted to oripavine, codeine, morphine, oxycodone, hydrocodone, oxymorphone, hydromorphone naltrexone, naloxone, hydroxycodone, neopinone, buprenorphine, or any combination thereof which each may subsequently be used to prepare a pharmaceutical formulation.

(230) The disclosure is now described with reference to the following Examples. These Examples are provided for the purpose of illustration only and the disclosure should in no way be construed as being limited to these Examples, but rather should be construed to encompass any and all variations which become evident as a result of the teaching provided herein.

EXAMPLES

Example 1—Preparation of Extracts from Plant Latex Containing Thebaine Synthesis Polypeptides

(231) Opium poppy plants were cultivated in growth chambers under a combination of fluorescent and incandescent lighting with a photoperiod of 16 h, and at 20° C./18° C. (light/dark).

(232) In order to prepare extracts containing thebaine synthesis polypeptides, a latex preparation was obtained from mature *Papaver somniferum* seedpods using the following procedure to obtain a purified plant extract containing thebaine synthesis polypeptide extract.

(233) A latex sample from mature *Papaver somniferum* seed capsules was extracted using phosphate buffer (Na—PO.sub.4), pH 7.0, containing 500 mM mannitol. The extract was centrifuged and the supernatant was recovered. The total soluble protein in the supernatant was desalted using a PD-10 desalting column consisting of Sephadex G-25 resin. Ammonium sulfate was added to the desalted protein sample to 40% (w/v) saturation. After centrifugation, additional ammonium sulfate was added to the supernatant to 80% (w/v) saturation. Subsequent centrifugation yielded a protein pellet, which was then resuspended in Na—PO.sub.4 buffer, pH 7.0, yielding the crude thebaine synthesis polypeptide extract. The protein extract was further purified using a series of chromatography methods. The sequential purification steps are described below.

(234) Following ammonium sulfate precipitation, the protein sample was loaded onto a butyl-S sepharose 6 Fast Flow column (12 mm×100 mm; GE Healthcare) equilibrated with 50 mM Na—PO.sub.4 buffer, pH 7.0, containing 1.7 M ammonium sulfate. Protein was eluted stepwise at 2 mL/min using 50 mM Na—PO.sub.4 buffer, pH 7.0, containing 1.0 M arginine-HCl, initially at 40% (v/v), then increasing to 79% (v/v) and finally reaching 100%. Elution fractions were collected and assayed for thebaine synthesis protein activity.

(235) Active fractions from butyl-S sepharose 6 chromatography were pooled, buffer exchanged to 20 mM Tris-HCl, pH 8.0, and loaded onto a 1.0 mL HiTrap Q HP column (GE Healthcare). Protein elution was performed with 20 mM Tris-HCl, pH 8.0, and 1.0 M NaCl applied with linear gradient of 0-60% (v/v) at a flow rate of 1 mL/min for 20 min. Elution fractions were collected and assayed for thebaine synthesis protein activity.

(236) Active fractions from HiTrap Q HP chromatography were concentrated and loaded onto two Superdex 75 HR 10/30 columns (GE Healthcare) connected in tandem. Isocratic elution was performed with 20 mM Tris-HCl, pH 8.0, 0.15 M NaCl and 10% (v/v) glycerol at a flow rate of 1 mL/min. Elution fractions were collected and assayed for thebaine synthesis protein activity.

(237) Active fractions from Superdex 75 HR 10/30 chromatography were pooled and applied to a Macro-Prep Ceramic Hydroxyapatite TYPE I column (BioRad) pre-equilibrated with 100 mM Na—PO.sub.4, pH 6.7, 90 mM NaCl and 20 mg/L CaCl.sub.2. Stepwise elution was performed using 500 mM Na—PO.sub.4, pH 6.7, containing 100 mM NaCl applied initially at 7% (v/v) for 7 mL, then increased to 10% (v/v) for 7 mL and finally reaching 100% for 15 mL at a flow rate of 1 mL/min.

(238) It was observed that when a *Papaver somniferum* latex preparation was extracted using a phosphate buffer without mannitol, thebaine synthesis polypeptide activity could not be detected. Extracts using a phosphate buffer with and without 500 mM mannitol were prepared and proteomically analyzed for the quantitative differential presence of individual polypeptides. The presence of relatively higher quantities in the proteome of a protein extract obtained following extraction using a phosphate buffer with mannitol was deemed indicative of a thebaine synthesis polypeptide. Several thebaine synthesis polypeptide candidates were identified. Accordingly, protein extracts comprising polypeptides having the sequences set forth in SEQ ID NO. 5; SEQ ID NO. 6; SEQ ID NO. 7; SEQ ID NO. 8; SEQ ID NO. 9; SEQ ID NO. 10; SEQ ID NO. 11; and SEQ ID NO. 12 resulted in a thebaine synthesis polypeptide mediated chemical reaction resulting in the conversion of substrate salutaridinol into thebaine. In particular, SEQ ID NO. 6 (known herein as Bet v1 (a.k.a., Betv1M/BetV1-1/THS2) showed a very high increase a thebaine synthesis polypeptide mediated chemical reaction.

(239) Thebaine synthesis polypeptide activity was measured by incubating protein extracts with salutaridinol and assaying for the presence of compounds having an m/z=312, corresponding with thebaine and compounds have an m/z=330, corresponding with morphinan alkaloid by-products, and further assaying for the possible presence of non-converted quantities of salutaridinol.

(240) Results are shown in FIGS. 7-8. In particular, FIG. 7 depicts certain relative amounts of thebaine produced by SalAT in the presence of several protein fractions (AS, HIC, IEC, HA, fractions) from the purification process this Example 1. Also shown is a polyacrylamide gel following gel electrophoresis of the same protein extracts and thus show thebaine synthesis protein activity present in these fractions. FIG. 7G depicts two polyacrylamide gels following gel electrophoresis of protein extracts (SEC and HA fraction) and purification of the 16-18 kDa bands after inclusion of the SEC step before the hydroxyapatite (HA) step, with retention of the thebaine synthesis protein (TS) activity.

(241) SalAT-coupled enzyme assays showed that less than 10% of the initial salutaridinol substrate was converted to thebaine (FIG. 6A). The major product was a compound, or a set of compounds, with m/z 330 that had not previously been reported in opium poppy (FIGS. 6B and 6C). The addition of a soluble protein extract from opium poppy latex to the SalAT-couples assays increased

thebaine formation at the expense of the m/z 330 compound (FIG. 6A). Interestingly, latex protein extracted with sodium phosphate buffer containing 500 mM mannitol resulted in substantially higher thebaine formation compared with latex protein extracted in the absence of mannitol.

(242) Opium poppy latex protein extracted from the high-thebaine and high-orphavine chemotype T was subjected to ammonium sulfate precipitation and, subsequently, a series of three different methods of chromatographic separation (i.e. hydrophobic-interaction [HIC] (FIG. 6D), ion-exchange [IEX] (FIG. 6E) and size exclusion [SEC] (FIG. 6F)). The partial purification resulted in a 20-fold increase in specific activity with respect to thebaine formation, and suggested a native protein molecular weight of approximately 40 kDa (FIG. 6G). Four successive SEC elution fractions were initially resolved on a 10% (w/v) SDS-PAGE gel as a single protein band of approximately 17 kDa (FIGS. 7B-7D). The protein samples were then pooled and the combined proteins resolved on a 14% (w/v) SDS-PAGE gel, which revealed three dominant bands of approximately 17 kDa (FIGS. 7B-7D). The three dominant protein bands were excised from the 14% (w/v) gel and analyzed individually by LC-MS/MS. Protein identification was performed using Scaffold (Proteome Software) proteomics software and a translated opium poppy latex transcriptome database. Over 100 proteins were detected in the three protein bands, yet the majority were represented by only a few spectral counts. The top six proteins showing the highest spectral counts (FIG. 7E) were produced individually in *E. coli* and *Saccharomyces cerevisiae* from codon-optimized genes. A protein sequence alignment showed that six candidates shared considerable amino acid sequence identity and all belong to the MLP- and Betv1-type PR10 protein family (FIG. 7F). SalAT-coupled enzyme assays were then performed with purified recombinant proteins. One of the six candidates (i.e., Betv1-1) substantially increased thebaine formation. The addition of 2 µg of the Betv1-1 (SEQ ID NO. 6) protein in a 50-4 SalAT-coupled reaction containing 20 µM salutaridinol as the substrate effectively eliminated production of the m/z-330 by-product (FIGS. 8A and 8B). Betv1 (SEQ ID NO. 6) alone (i.e. without SalAT in the assay) did not transform salutaridinol into a product. In yeast containing an salutaridine reductase (SalR) and SalAT gene and, individually, each of the six candidates transiently expressed from a plasmid, only Betv1-1 (SEQ ID NO. 6) facilitated the enhanced conversion of exogenous salutaridine to the thebaine (FIGS. 9A-9C). Betv1-1 (SEQ ID NO. 6) was designated as the primary thebaine synthase (THS).

Example 2—Enzymatic Production of Thebaine In Vitro

(243) Bet v1-1A (corresponding to C-terminal HA tagged), Bet v1-1B (corresponding to N-terminal HA tagged), PR10-3A (SEQ ID NO. 33), and PR10-3B (SEQ ID NO. 34) were expressed in *E. coli* (Rosetta). Bet v1 corresponds to SEQ ID NO. 6. The transformants were used to produce crude extracts and tested for thebaine synthesis activity. Extracts for each expression construct were prepared and used to prepare various thebaine assay mixtures, as shown in Table 1.

(244) TABLE-US-00001

TABLE 1	Bet v1-	Bet v1-	PR10-	"All-in-	SalAT	Salutaridinol	Acetyl-CoA	Phosphate	buffer	pH7	Crude <i>E. coli</i>	extract
One"	Purified	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

(245) The assay mixtures were used to determine the presence thebaine and m/z 330 (an unknown product).

(246) Extracts from *E. coli* expressing Bet v1-1B produced the most thebaine whereas Bet v1-1A produced slightly less. Extracts from *E. coli* expressing SaltAT, Bet v1-1A, Bet v1-1B, PR10-3A and PR10-3B, produced the highest titers of thebaine. FIG. 9D.

(247) In the assays using thebaine synthesis polypeptide candidates in *E. coli* extracts, the majority of salutaridinol was not consumed, whereas with SalAT alone, all salutaridinol was used and formed m/z 330 byproduct. FIG. 9D.

(248) Bet v1-1A and Bet v1-1B produce thebaine and no m/z 330 byproduct. PR10-3A and PR10-3B produce little to no thebaine, but also had no m/z 330 byproduct formation. In cases of mixtures containing Bet v1-1A and Bet v1-1B or PR10-3A and PR10-3B, consumption of salutaridinol was inhibited (FIG. 10). With SalAT alone, all salutaridinol was converted mostly to m/z 330 (FIG. 10)

and very little to no thebaine (FIG. 9D).

Example 3—the Thebaine Synthase Cluster

(249) A draft genomic DNA sequencing of opium poppy chemotype Roxanne provided information about the genomic structural relationship between THS and other thebaine biosynthetic genes (FIG. 11A). On a genomic scaffold of approximately 1 megabase pairs duplicate copies for each of 5 thebaine biosynthetic genes were physically linked. The five gene products, reticuline epimerase (REPI), salutaridine synthase (SalSyn), SalR, SalAT, and THS, sequentially convert (S)-reticuline to thebaine. The duplication of each of the five genes in a single cluster suggests the occurrence of complex genomic rearrangement to coordinate and enhance the expression of thebaine biosynthetic genes. The two THS genes encode two highly similar yet distinct THS isoforms, one (THS1; SEQ ID NO. 80) composed of 181 amino acids and the other (THS2; SEQ ID NO. 6) containing 163 amino acids (FIG. 11B). A conserved cryptic downstream start codon in both THS1 and THS2 yield N-terminally truncated versions of each protein, designated THS1s (131 amino acids; SEQ ID NO. 81) and THS2s (133 amino acids; SEQ ID NO. 32). An additional 12 amino acids at the N-terminus of THS1, compared with THS2, is predicted to encode a secretion signal peptide. Moreover, compared with THS2, THS2s results from alternate splicing that the first ~130 base pairs at the 5' end (including both UTR and ORF regions) of THS2, which are missing in the corresponding transcript. Compared with THS2, four additional nucleotides near the start of the THS1 transcript result in an upstream translation start site. Alternate splicing in THS1 also yields the shorter THS product.

(250) As a member of the PR10 family, opium poppy THS and orthologs from other thebaine-producing *Papaver* species are divergent from other plant PR10 proteins including norcoclaurine synthase (NCS), which catalyzes the first committed step of BIA biosynthesis (FIG. 11D). Tissue-specific expression analysis was performed by qRT-PCR using gene- and splice variant-specific primers (FIG. 12A and FIG. 12B) on two opium poppy chemotypes, T (a high-thebaine, high-orphavine and no-morphine, no-codeine variety) and Roxanne (a variety producing morphine, codeine, as well as noscapine and papaverine). Similar to the expression profile of genes encoding other late morphine pathway enzymes, THS transcripts were more abundant in latex, stem and capsule (FIG. 12C). In general, the truncated transcripts encoding THS1s (SEQ ID NO. 81) and THS2s (SEQ ID NO. 32) were expressed at higher level compared with the longer THS1 (SEQ ID NO. 80) and THS2 (SEQ ID NO. 6) versions.

(251) Suppression of THS transcript levels in opium poppy plants was performed using virus-induced gene silencing (VIGS). Conserved sequences of the THS1 and THS2 genes were used as the target region (FIG. 12A). THS transcript levels were significantly lower in THS-silenced plants compared with controls in plants infected with *Agrobacterium tumefaciens* harboring pTRV2-THS and pTRV2 empty-vector plasmids, respectively (FIG. 13A and FIG. 13B). Thebaine content in the THS-silenced plants was significantly reduced nearly two-fold compared with controls FIG. 13B). In contrast, accumulation of the upstream intermediates reticuline and salutaridinol increased significantly in THS-silenced compared with control plants. Interestingly, morphine and codeine levels were not significantly affected suggesting that the reduced THS activity was still able to support sufficient thebaine production to enable final product formation.

(252) All versions of THS1 and THS2 transcripts were codon-optimized and expressed in *E. coli*. Purified recombinant THS proteins were added to SalAT-coupled enzyme assays. Both THS1 and THS2 assays similarly increased thebaine formation, whereas the shorter versions did not show catalytic activity (FIG. 14B). Size exclusion chromatography (SEC) during the purification of THS from opium poppy latex revealed a molecular weight of ~40 kDa, suggesting that the native protein occurs as a homodimer. Chemical cross-linking with the purified, recombinant THS proteins was performed to confirm dimerization (FIGS. 15A and 15B). Both THS1 and THS2 can form homodimers, and also display the formation of higher molecular weight oligomers at lower abundance. In contrast, the shorter THS1s and THS2s versions exhibited considerably a lower

ability to form dimers, but did yield higher molecular weight oligomers. Subjecting the native THS2 protein to SEC chromatography showed a major peak of ~40 kDa with lesser quantities of higher molecular weight peaks (FIGS. 15A and 15B). To further investigate the relative abundance of THS isoforms in the plant, protein extracts derived from whole stem, latex-free stem, and latex isolated from the T chemotype were subjected to proteomics analysis (FIG. 11C). In detected peptides with at least one amino acid substitution spectral counts for THS2 were considerably higher than for THS1, consistent with the higher abundance of THS2, compared with THS1, transcripts (FIG. 12C).

(253) The catalytic function of THS1 and THS2 was investigated using salutaridinol 7-O-acetate as a direct enzymatic substrate. Salutaridinol 7-O-acetate is the immediate product of SalAT reaction and is normally not detectable because it spontaneously converts to a small amount of thebaine and a larger amount of m/z 330 compound(s). Approximately 70% of the salutaridinol 7-O-acetate used as a substrate for assays broke down within 2 min in aqueous solution; thus, THS assays using salutaridinol 7-O-acetate as the substrate were performed with an optimized quality of purified, recombinant THS2 for 30 sec, which was within the linear range of product formation. Boiled enzyme served as the control in all enzymatic assays to compensate for any spontaneous breakdown of salutaridinol 7-O-acetate. Chloroform was used as the storage solution for salutaridinol 7-O-acetate and quenching solution for the reaction, which prevented further degradation. THS2 showed no activity with other benzyloisoquinoline alkaloids either sharing similar skeleton or containing an O-acetyl moiety (FIG. 16A). Competition assays using these compounds revealed marginal inhibition of THS activity associated with salutaridine and salutaridinol, but little if any effect of O-acetylated compounds (FIG. 16B).

(254) Inclusion of THS2 in engineered yeast with chromosomally integrated biosynthetic genes facilitating the conversion of (R,S)-norlauranosoline to salutaridinol 7-O-acetate (Sc-3) substantially increased the yield of thebaine from (R,S)-norlauranosoline (FIGS. 18A-18C, FIGS. 21A-21C). A strain (Sc-2) capable only of salutaridine production, owing to the lack of genes encoding SalR and SalAT, was unable to produce thebaine. Episomal expression of THS2 in a strain (Sc-4) containing a chromosomally integrated THS2 gene further enhanced thebaine formation to ~750 µg/L. The m/z 330 by-product decreased as thebaine production increased.

Example 4—the Effect of HA Tag on Bet v1-1 Thebaine Synthesis Activity In Vivo

(255) In order to test whether the HA-tag had an effect on the Bet v1-1 (SEQ ID NO. 6) efficacy and efficiency, 2 constructs (N- and C-terminal HA-tagged Bet v1-1) were expressed on high copy plasmids and expressed in yeast. The transformed yeast were fed with 1 mM salutaridine for 48 hours and tested for its capability to make thebaine. As shown in FIG. 17, both the N- and C-terminal HA-tagged Bet v1-1 constructs had thebaine synthesis activity, however, the N-terminal tagged Bet v1-1 showed a significant increase in thebaine synthesis activity.

Example 5—Enzymatic Production of Thebaine in Yeast

(256) Three separate yeast strains were generated having the genotype in Table 2:

(257) TABLE-US-00002 TABLE 2 Strain Name Genotype Sc2 CEN.PK

T.sub.MET13::T.sub.CBS1 P.sub.GAL1-Pbra6OMT-T.sub.ADH1 T.sub.LYS2:: T.sub.CST1
P.sub.GAL10-Psom4OMT-T.sub.PMA1 P.sub.GAL1-CjapCNMT- T.sub.PGK1
T.sub.LYS4::T.sub.CHS2 P.sub.GAL10 -PsoREPI-2-T.sub.ADH1 T.sub.MET17::T.sub.TUB1
P.sub.GAL10-PbrSalSyn-T.sub.ADE3 P.sub.GAL1-PbrCPR1-T.sub.PDR5 Sc3 Sc2
T.sub.MET16::T.sub.CHS2 P.sub.GAL10-PsomSalR-T.sub.HPC2 P.sub.TEA1-PsomSalAT-
T.sub.ROX3 Sc4 Sc3 T.sub.NCL1::T.sub.STE5P.sub.GAL1-PsomBetv1-1-T.sub.ADH1

Yeast were grown in medium supplemented with 2.5 mM NLDS and 10 mM L-methionine for 96 hours. Thebaine titers of the supernatants of three engineered yeast strains (SC-2, SC-3 and SC-4) were measured. As seen in FIG. 18A, SC-2 produced no detectable levels of thebaine. Further, SC-3 produced increased amount of thebaine in the presence of integrated HA-BETv1-1 (N-terminal HA tagged version of SEQ ID NO. 6). SC-4 showed higher expression when both integrated copies

and plasmid expressed HA-BETv1-1 were present. As shown in FIG. 18B, relative peak area of unknown side product (m/z 330) decreased in the supernatant in SC-3 and SC-4 strains in the presence of HA-BETv1-1. Error bars indicate standard deviations from four biological replicates. (258) Yeast expressing pGAL SalR and pTEA1 SalAT were also transformed with different thebaine synthesis polypeptide expression constructs. See Table 3. The thebaine synthesis polypeptide constructs were cloned into a high copy vector under pGAL. HA epitope tags were also present on both the N- and C-terminus.

(259) TABLE-US-00003 TABLE 3 Annotation Accession number Molecular weight (kDa) PR10-3 c25055_g1_i1 18 Betv1-1 c15408_g1_i1 18 PR10-5 c38417_g1_i1 17 PR10-4 c50593_g1_i1 18 MLP15 c27108_g1_i1 18 PR10-7 c33864_g1_i1 18 MLP-2 c37788_g2_i4 16 MLP-3 c1643_g1_i1 16

(260) The cells were cultured in SC-Complete Media with 100 mM MES pH 7.5 (G418 selection) and were fed with 50 μ M Salutaridine for 48 hours. After 48 hours, relative thebaine titers were measured. Relative titers for Bet v1-1 (SEQ ID NO. 6) were the highest (Data shown in FIG. 19A). N-terminal tagged Bet v1-1 exhibited an approximate 17-fold increase in thebaine titers (compared to empty vector controls) whereas C-terminal tagged Bet v1-1 produced an approximate 5-fold increase. Additionally, the unknown side product m/z 330 was also measured. As shown in FIG. 19B, the peak area of m/z 300 is significantly lower in yeast expressing the N-terminal tagged Bet v1-1. Peak area is shown relative to the empty vector control. Error bars indicate standard deviations from three biological replicates.

Example 6—Effect of Integration on Thebaine Synthesis

(261) Yeast expressing a stably integrated copy of Bet v1-1 (SEQ ID NO. 6) was created and tested for its ability to synthesize thebaine. The transformed yeast was cultured in media supplemented with 1 mM salutaridine for 48 hours. Integrated yeast showed an approximate 6-fold increase in thebaine synthesis activity. See FIG. 12A.

(262) Yeast expressing stably integrated copies of Bet v1-1 splice variants (Bet v1-1S, and Bet v1-1L) were created and tested for its ability to synthesize thebaine. See Table 4 and FIG. 20C for sequences:

(263) TABLE-US-00004 TABLE 4 SEQ. ID. NO. NAME AMINO ACID SEQUENCE 6
BET V1-1 MAPLGVSGLVGKLSTELEVDC EDAKYNNMYKHGEDVKKAVPH
LCVDVKIISGDPTSSGCIKEW NVNIDGKTIRSVEETTHDDE TKTLRHRVFEGDVMKDFKKF
DTIMVVNPKPDGNGC VVTRS IEYEKTNENSPTPFDYLQFG HQAIEDMNKYLRDSESN 31
BET V1-1L MDSINSSIFYCAYFRELIK LLAPLGVSGLVGKLSTELE
VDCDAEKYNNMYKHGEDVKK AVPHLCVDVKIISGDPTSSG CIKEWNVNIDGKTIRSVEET
THDDETKTLRHRVFEGDVMK DFKKFDTIMVVNPKPDGNGC
VVTRSIEYEKTNENSPTPFD YLQFGHQ AIEDMNKYLRDSE 32 BET V1-1S
MYKHGEDVKKAVPHLCVDVK IISGDPTSSGCIKEWNVNID GKTIRSVEETTHDDETKTLR
HRVFEGDVMKDFKKFDTIMV VNP KPDGNGC VVTRSIEYEK TNENSPTPFDYLQFGHQ AIE
DMNKYLRDSESN

(264) The transformed yeasts were cultured in media supplemented with 1 mM salutaridine for 48 hours. Yeast expressing the splice variant Bet v1-1S (SEQ ID NO. 32) did not exhibit any thebaine synthesis activity. See FIG. 20B. Yeast expressing the splice variant Bet v1-1L (SEQ ID NO. 31) exhibited similar, if not higher, thebaine synthesis activity.

Example 7—Thebaine, Reticuline and Salutaridine Titers

(265) The strains described in Table 2 were tested for ability to produce thebaine, reticuline, and salutaridine. For each strain tested, at least three individual colonies were picked and placed into 0.8 mL of Synthetic Complete-Monosodium Glutamate (SC-MSG) media containing 2% glucose and 200 mg/L of G418 in 96-deep well plates (Axygen Scientific). Cultures were grown in a Multitron Pro shaker (Infors HT) at 30° C. with 80% humidity.

(266) As shown in FIG. 21A, strain SC-2 did not produce any detectable levels of thebaine at 24,

38 or 96 hours. Strain SC-3 produced significantly increased amounts of thebaine between 24 and 48 hours. Strain SC-4 produced the highest thebaine titers of all strains tested and produced significantly increased amounts of thebaine between 24 and 48 hours.

(267) All strains tested produced significant levels of reticuline titers. (See FIG. 21B) All strains also exhibited an increased production of reticuline between 24, 48 and 96 hours.

(268) As shown in FIG. 21C, strain SC-2 produced the most amount of salutaridine comprised to strains SC-3 and SC-4. Generally, the strains also exhibited an increased production of salutaridine between 24, 48 and 96 hours.

Example 8—Purine Permeases Effect on Thebaine and Other BIA Intermediates

(269) A *Saccharomyces cerevisiae* strain were transformed with one of the 11 different following purine permease homologs: 1) *Arabidopsis thaliana* PUP 1 (AtPUP1; SEQ ID NO: 41 (amino acid) and 42 (nucleotide)); 2) *Jatropha curcas* PUP1 (JcurPUP1 (JC_PUP1; SEQ ID NO: 43 (amino acid) and 44 (nucleotide)); 3) *Papaver somniferum* PUP1-1 (PsoPUP1-1; SEQ ID NO: 45 (amino acid) and 46 (nucleotide)); 4) *Papaver somniferum* PUP1-2 (PsoPup1-2; SEQ ID NO: 47 (amino acid) and 48 (nucleotide)); 5) *Papaver somniferum* PUP1-3 (PsoPUP1-3; SEQ ID NO: 49 (amino acid) and 50 (nucleotide)); 6) *Papaver alpinum* PUP1 (PalpPUP1; SEQ ID NO: 53 (amino acid) and 54 (nucleotide)); 7) *Papaver atlanticum* PUP1 (PatlPUP1; SEQ ID NO: 55 (amino acid) and 56 (nucleotide)); 8) *Papaver myabeanum* PUP1 (PmiyPUP1; SEQ ID NO: 57 (amino acid) and 58 (nucleotide)); 9) *Papaver nudicale* PUP1 (PnudPUP1; SEQ ID NO: 59 (amino acid) and 60 (nucleotide)); 10) *Papaver radicum* PUP1 (PradPUP1; SEQ ID NO: 61 (amino acid) and 62 (nucleotide)); and 11) *Papaver trinifolium* PUP1 (PtriPUP1; SEQ ID NO: 63 (amino acid) and 64 (nucleotide)). The strains were grown in media supplemented with 2.5 nM NLDS and 1 mM methionine for 24 and 48 hours. Titers of respective end products were measured and OD adjusted. As seen in FIG. 23A, S-reticuline levels were generally higher across the board for those strains that expressed a purine permease. Similarly, as seen in FIG. 23B, R-reticuline levels were generally higher across the board for those strains that expressed a purine permease. As seen in FIG. 23C, salutaridine levels were increased in the presence of PsoPUP1-1 and PsoPUP1-3 purine permease. In the presence of an L-DOPA feed, as seen in FIG. 23D, NLDS levels remained largely unchanged whereas reticuline levels were generally increased in the presence of purine permeases.

(270) Strains were transformed with one or more *Papaver somniferum* purine permeases and thebaine, salutaridinol, and salutaridine titers were measured. The strains were all transformed with BETV1 (SEQ ID NO. 6) in combination with one or more purine permeases for 48 hours. A nucleotide sequence encoding for BetV1 was integrated into the genome of the yeast strain. Further, a purine permease (PUP-1—SEQ ID NO 46)) was integrated into the genome of the yeast strain (control—left side). A PsomPUP1-4 (SEQ ID NO. 51 (amino acid) and 52 (nucleotide)) was expressed via a high copy plasmid (second from left). Additionally, JcurPUP1 (SEQ ID NO: 43 (amino acid) and 44 (nucleotide)) was integrated into the control strain, thus expressing both integrated copies of JcurPUP1 and PsomPUP1 (control—third from the left). Finally, three copies of a purine permease were expressed, PsomPUP1 integrated, JcurPUP1 integrated, and high copy plasmid expressing PUP1-4 (far right). As seen in FIG. 24, yeast strains that expressed a plasmid copy of PsomPUP1-4 (SEQ ID NO. 52), plus an integrated copy of PsomPUP1-1 (SEQ ID NO. 46), and BetV1, produced significantly high levels of thebaine titers, compared to controls not expressing PsomPUP1-4. However, when PsomPUP1-1, PsomPUP1-4, and JcurPUP1 were all expressed in a yeast strain, thebaine production significantly decreased, and were similar to expressing a single PsomPUP1-1. Significantly, when three purine permeases were expressed, overall carbon flow to thebaine, salutaridinol, and salutaridine, decreased approximately 50%.

Example 9—Expression of Betv-1 in Yeast Expressing SalAT and SalR Genes, and Fed Salutaridine, and Co-Expression of Betv-1 and Purine Permease in Yeast Expressing SalAT and SalR Genes, and Fed Salutaridine

(271) A *Saccharomyces cerevisiae* strain CENPK102-5B expressing salutaridinol 7-O-

acetyltransferase (SalAT) and salutaridine reductase (SalR) genes integrated into the yeast genome was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing Betv-1 (SEQ ID NO: 6) and purine permease polypeptides as follows: (i) a nucleic acid sequence encoding Betv-1 alone (SEQ. ID NO: 6) (HA_BetV1M); (ii) a nucleic acid sequence encoding Betv-1 alone (SEQ. ID NO: 31) (BETV1L_HA); a nucleic acid sequence expressing Betv-1 (SEQ. ID NO: 6) and purine permease (SEQ. ID NO: 35) (HA_BetV1M-pp); and (iv) a nucleic acid sequence expressing Betv-1 (SEQ. ID NO: 31) and purine permease (SEQ. ID NO: 35) (BetV1L_HA-pp). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of 100 μ M salutaridine for 24 hrs. An aliquot of 5 μ L of culture medium was subjected to mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the thebaine concentration in the medium of each strain was determined according to a thebaine standard curve. The results are shown in FIG. 25. As can be seen in FIG. 25, thebaine production increased 6-fold (to approximately 600 μ g/L/OD) in the medium containing strains transformed with only a Betv-1 (HA_BetV1M and BETV1L_HA) compared with strains expressing only SalAT and SalR. However, and surprisingly, in excess of 9,000 μ g/L/OD thebaine was detected in the medium containing strains transformed with both Betv_1 and purine permease (HA_BetV1M-pp and BetV1L_HA-pp), thus, generating an additional 16-fold increase compared with strains with only a Betv-1 (HA_BetV1M and BETV1L_HA). The production of thebaine increased approximately 100-fold in strains transformed with both Betv_1 and purine permease (HA_BetV1M-pp and BetV1L_HA-pp) compared with strains expressing only SalAT and SalR.

Example 10—Expression of Betv-1 in Yeast and Co-Expression of Betv-1 and Purine Permease in Yeast, Fed Salutaridine

(272) A *Saccharomyces cerevisiae* strain CENPK102-5B was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing Betv-1 and purine permease polypeptides as follows: (i) a nucleic acid sequence encoding Betv-1 alone (SEQ. ID NO: 6) (HA_BetV1M); (ii) a nucleic acid sequence encoding Betv-1 alone (SEQ. ID NO: 31) (BETV1L_HA); a nucleic acid sequence expressing Betv-1 (SEQ. ID NO: 6) and purine permease (SEQ. ID NO: 35) (HA_BetV1M-pp); and (iv) a nucleic acid sequence expressing Betv-1 (SEQ. ID NO: 31) and purine permease (SEQ. ID NO: 35) (BetV1L_HA-pp). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of 100 μ M salutaridine for 24 hrs. Yeast cells were collected by centrifugation, extracted in 500 μ L of methanol, of which 5 μ L was subjected to mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the salutaridine concentration in the cells of each strain was determined according to a salutaridine standard curve. The results are shown in FIG. 26. As can be seen in FIG. 26, salutaridine accumulation increased 8-fold (to approximately 132 μ g/L/OD) in the cells of strains transformed with a Betv-1 (HA_BetV1M and BETV1L_HA) and a purine permease (HA_BetV1M-pp and HA_BetV1L-pp). Compared with the empty vector (EV) control, salutaridine accumulation did not increase in cells transformed only with a Betv-1 (HA_BetV1M and BETV1L_HA).

Example 11—Expression of Purine Permeases (PUP-L) and (PUP-N) in Yeast Expressing DODC, MAO, NCS, 6OMT, CNMT and 4'OMT Genes, and Fed Either L-DOPA or Norlaudanosoline (NLDS), and Co-Expression of PUP-L and PUP-N with a Betv1 in Yeast Expressing DODC, MAO, NCS, 6OMT, CNMT and 4'OMT Genes, and Fed Either DOPA or Norlaudanosoline (NLDS)

(273) A *Saccharomyces cerevisiae* strain CENPK102-5B expressing dopa decarboxylase (DODC), monoamine oxidase (MAO), norcoclaurine synthase (NCS), norcoclaurine 6-O-methyltransferase (6OMT), coclaurine N-methyltransferase (CNMT), and 3'hydroxy-N-methyltransferase 4'-O-

methyltransferase (4'OMT) genes integrated into the yeast genome was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing purine permease and Betv1 polypeptides as follows: (i) a nucleic acid sequence encoding a purine permease containing a C-terminal extension absent in a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer, T., Gazda, V., He Z., Kaminski F., Kern M., Larson T. R., Li Y., Meade F., Teodor R., Vaistij F. E., Walker C., Bowser T. A., Graham, I. A. (2012) *A Papaver somniferum* 10-gene cluster for synthesis of the anticancer alkaloid noscapine. Science 336(6089): 1704-1708) alone (SEQ. ID NO: 35) (PUP-L); (ii) a nucleic acid sequence encoding a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 37) (PUP-N) (nucleic acid sequence SEQ. ID NO: 38 was codon optimized and C-terminally myc-tagged to obtain nucleic acid sequence SEQ. ID NO: 39 (myc-tag sequence SEQ. ID NO: 40)); (iii) a nucleic acid sequence expressing PUP-L (SEQ. ID NO: 35) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-L); and (iv) a nucleic acid sequence expressing PUP-N (SEQ. ID NO: 37) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-N). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of either 100 μ L-DOPA or 100 μ M norlaudanoline (NLDS) for 24 hrs. An aliquot of 5 μ L of culture medium was subjected to mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the reticuline concentration in the medium of each strain was determined according to a reticuline standard curve. The results are shown in FIG. 27. As can be seen in FIG. 27, reticuline production was not affected in the medium of strains transformed with only a purine permease (PUP-L or PUP-N) or a purine permease and a Betv1 (Betv1 PUP-L and Betv1 PUP-N) compared with strains expressing only DODC, MAO, NCS, 6OMT, CNMT and 4'OMT. However, and surprisingly, reticuline production increased 25-fold to more than 1,000 μ g/L/OD in the medium containing strains transformed with both PUP-L and PUP-L and Betv1. Neither PUP-N nor Betv1 alone affected the production of reticuline compared with empty vector controls.

Example 12—Expression of Purine Permeases (PUP-L) and (PUP-N) in Yeast Expressing REPI, CPR, and SalSyn Genes or REPI, CPR, SalSyn, SalR and SalAT Genes, and Fed (S)-Reticuline, and Co-Expression of PUP-L and PUP-N with a Betv1

(274) A *Saccharomyces cerevisiae* strain CENPK102-5B expressing reticuline epimerase (REPI), cytochrome P450 reductase (CPR) and salutaridine synthase (SalSyn) genes integrated into the yeast genome was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing purine permease and Betv1 polypeptides as follows: (i) a nucleic acid sequence encoding a purine permease containing a C-terminal extension absent in a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 35) (PUP-L); (ii) a nucleic acid sequence encoding a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 37) (PUP-N); (iii) a nucleic acid sequence expressing PUP-L (SEQ. ID NO: 35) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-L); and (iv) a nucleic acid sequence expressing PUP-N (SEQ. ID NO: 37) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-N). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of 100 μ M (S)-reticuline for 24 hrs. An aliquot of 5 μ L of culture medium was subjected to mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the salutaridine and thebaine concentrations in the medium of each strain were determined according to salutaridine and thebaine standard curves, respectively. The results are shown in FIG. 28A. As can be seen in FIG. 28A, salutaridine production was not affected in the medium of strains transformed with PUP-N, or PUP-N and a Betv1 (Betv1 PUP-N) compared with strains expressing only REPI, CPR, and SalSyn. However, and surprisingly, salutaridine production increased 3-fold compared with the empty vector (EV) control to more than 150 μ g/L/OD in the medium containing strains

transformed with either PUP-L alone or PUP-L and a Betv1 (Betv1 PUP-L). B, A *Saccharomyces cerevisiae* strain CENPK102-5B expressing reticuline epimerase (REPI), cytochrome P450 reductase (CPR), salutaridine synthase (SalSyn), salutaridine reductase (SalR), and salutaridine acetyltransferase (SalAT) genes integrated into the yeast genome was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing purine permease and Betv1 polypeptides as follows: (i) a nucleic acid sequence encoding a purine permease containing a C-terminal extension absent in a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 35) (PUP-L); (ii) a nucleic acid sequence encoding a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 37) (PUP-N); (iii) a nucleic acid sequence expressing PUP-L (SEQ. ID NO: 35) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-L); and (iv) a nucleic acid sequence expressing PUP-N (SEQ. ID NO: 37) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-N). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of 100 μ M (S)-reticuline for 24 hrs. An aliquot of 5 μ L of culture medium was subjected to mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the salutaridine and thebaine concentrations in the medium of each strain were determined according to salutaridine and thebaine standard curves, respectively. The results are shown in FIG. 28B. As can be seen in FIG. 28B, salutaridine and thebaine production was not affected in the medium of strains transformed with PUP-N compared with strains expressing only REPI, CPR, SalSyn, SalR and SalAT. However, and surprisingly, salutaridine production increased 2-fold compared with the empty vector (EV) control to approximately 150 μ g/L/OD in the medium containing strains transformed with either PUP-L alone or PUP-L and a Betv1 (Betv1 PUP-L). Thebaine production increased 4-fold compared with the empty vector (EV) control in the medium containing strains transformed with PUP-L alone. Thebaine production increased almost 20-fold compared with the empty vector (EV) control to approximately 95 μ g/L/OD in the medium containing strains transformed with PUP-L and a Betv1 (Betv1 PUP-L). Thebaine production increased approximately 2-fold compared with the empty vector (EV) control to approximately 12 μ g/L/OD in the medium containing strains transformed with PUP-N and a Betv1 (Betv1 PUP-L) owing to the thebaine-forming activity of Betv1.

Example 13—Expression of a Purine Permeases (PUP-L) and (PUP-N) in Yeast Expressing REPI, CPR, and SalSyn Genes or REPI, CPR, SalSyn, SalR and SalAT Genes, and Fed (R)-Reticuline, and Co-Expression of PUP-L and PUP-N with Betv1

(275) A *Saccharomyces cerevisiae* strain CENPK102-5B expressing reticuline epimerase (REPI), cytochrome P450 reductase (CPR) and salutaridine synthase (SalSyn) genes integrated into the yeast genome was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing purine permease and Betv1 polypeptides as follows: (i) a nucleic acid sequence encoding a purine permease containing a C-terminal extension absent in a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 35) (PUP-L); (ii) a nucleic acid sequence encoding a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 37) (PUP-N); (iii) a nucleic acid sequence expressing PUP-L (SEQ. ID NO: 35) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-L); and (iv) a nucleic acid sequence expressing PUP-N (SEQ. ID NO: 37) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-N). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of 100 μ M (R)-reticuline for 24 hrs. An aliquot of 5 μ L of culture medium was subjected to mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the salutaridine and thebaine concentrations in the medium of each strain were determined according to salutaridine and thebaine standard curves, respectively. The results are shown in FIG. 29A. As can be seen in FIG. 29A,

salutaridine production was not affected in the medium of strains transformed with PUP-N, or PUP-N and a Betv1 (Betv1 PUP-N) compared with strains expressing only REPI, CPR, and SalSyn. However, and surprisingly, salutaridine production increased 3-fold compared with the empty vector (EV) control to more than 150 µg/L/OD in the medium containing strains transformed with either PUP-L alone or PUP-L and a Betv1 (Betv1 PUP-L). B, A *Saccharomyces cerevisiae* strain CENPK102-5B expressing reticuline epimerase (REPI), cytochrome P450 reductase (CPR), salutaridine synthase (SalSyn), salutaridine reductase (SalR), and salutaridine acetyltransferase (SalAT) genes integrated into the yeast genome was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing purine permease and Betv1 polypeptides as follows: (i) a nucleic acid sequence encoding a purine permease containing a C-terminal extension absent in a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 35) (PUP-L); (ii) a nucleic acid sequence encoding a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 37) (PUP-N); (iii) a nucleic acid sequence expressing PUP-L (SEQ. ID NO: 35) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-L); and (iv) a nucleic acid sequence expressing PUP-N (SEQ. ID NO: 37) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-N). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of 100 µM (R)-reticuline for 24 hrs. An aliquot of 5 µL of culture medium was subjected to mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the salutaridine and thebaine concentrations in the medium of each strain were determined according to salutaridine and thebaine standard curves, respectively. The results are shown in FIG. 29B. As can be seen in FIG. 29B, salutaridine and thebaine production was not affected in the medium of strains transformed with PUP-N compared with strains expressing only REPI, CPR, SalSyn, SalR and SalAT. However, and surprisingly, salutaridine production increased 2-fold compared with the empty vector (EV) control to approximately 150 µg/L/OD in the medium containing strains transformed with either PUP-L alone or PUP-L and a Betv1 (Betv1 PUP-L). Thebaine production increased 4-fold compared with the empty vector (EV) control in the medium containing strains transformed with PUP-L alone. Thebaine production increased almost 20-fold compared with the empty vector (EV) control to approximately 95 µg/L/OD in the medium containing strains transformed with PUP-L and a Betv1 (Betv1 PUP-L). Thebaine production increased approximately 2-fold compared with the empty vector (EV) control to approximately 12 µg/L/OD in the medium containing strains transformed with PUP-N and a Betv1 (Betv1 PUP-N) owing to the thebaine-forming activity of Betv1.

Example 14—Expression of Purine Permeases (PUP-L) and (PUP-N) in Yeast Expressing SalAT and SalR Genes or REPI, CPR, SalSyn, SalR and SalAT Genes, and Fed Salutaridine and Co-Expression of PUP-L and PUP-N with a Betv1

(276) A *Saccharomyces cerevisiae* strain CENPK102-5B expressing salutaridine reductase (SalR) and salutaridine acetyltransferase (SalAT) genes integrated into the yeast genome was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing purine permease and Betv1 polypeptides as follows: (i) a nucleic acid sequence encoding a purine permease containing a C-terminal extension absent in a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 35) (PUP-L); (ii) a nucleic acid sequence encoding a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 37) (PUP-N); (iii) a nucleic acid sequence expressing PUP-L (SEQ. ID NO: 35) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-L); and (iv) a nucleic acid sequence expressing PUP-N (SEQ. ID NO: 37) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-N). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of 100 µM salutaridine for 24 hrs. An aliquot of 5 µL of culture medium was subjected to

mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the thebaine concentration in the medium of each strain was determined according to a thebaine standard curve. The results are shown in FIG. 30A. As can be seen in FIG. 30A, thebaine production was not affected in the medium of strains transformed with PUP-N compared with strains expressing only SalR and SalAT. However, and surprisingly, thebaine production increased 27-fold and 60-fold compared with the empty vector (EV) control to more than 1600 and 3500 $\mu\text{g/L/OD}$ in the medium containing strains transformed with either PUP-L alone or PUP-L and a Betv1 (Betv1 PUP-L), respectively. Thebaine production increased approximately 4-fold compared with the empty vector (EV) control to approximately 250 $\mu\text{g/L/OD}$ in the medium containing strains transformed with PUP-N and a Betv1 (Betv1 PUP-N) owing to the thebaine-forming activity of Betv1. B, A *Saccharomyces cerevisiae* strain CENPK102-5B expressing reticuline epimerase (REPO, cytochrome P450 reductase (CPR), salutaridine synthase (SalSyn), salutaridine reductase (SalR), and salutaridine acetyltransferase (SalAT) genes integrated into the yeast genome was transformed with a yeast expression vector pEV-1 harboring various polynucleotide constructs expressing purine permease and Betv1 polypeptides as follows: (i) a nucleic acid sequence encoding a purine permease containing a C-terminal extension absent in a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 35) (PUP-L); (ii) a nucleic acid sequence encoding a purine permease linked to a cluster of 10 noscapine biosynthetic genes (Winzer et al., 2012) alone (SEQ. ID NO: 37) (PUP-N); (iii) a nucleic acid sequence expressing PUP-L (SEQ. ID NO: 35) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-L); and (iv) a nucleic acid sequence expressing PUP-N (SEQ. ID NO: 37) and a Betv1 (SEQ. ID NO: 6) (Betv1-PUP-N). The strains, as well as a control strain with pEV-1 not comprising a nucleic acid sequences encoding Betv-1 or purine permease (EV) were separately cultivated in growth medium SD-Leu-His in the presence of 100 μM salutaridine for 24 hrs. An aliquot of 5 μL of culture medium was subjected to mass spectrometry analysis using an LTQ-Orbitrap XL high-resolution mass spectrometer. Thereafter the thebaine concentration in the medium of each strain was determined according to a thebaine standard curve. The results are shown in FIG. 30B. As can be seen in FIG. 30B, thebaine production was not affected in the medium of strains transformed with PUP-N compared with strains expressing only REPI, CPR, SalSyn, SalR and SalAT. However, and surprisingly, thebaine production increased 10-fold and 25-fold compared with the empty vector (EV) control to approximately 1000 and 2500 $\mu\text{g/L/OD}$ in the medium containing strains transformed with either PUP-L alone or PUP-L and a Betv1 (Betv1 PUP-L), respectively. Thebaine production increased approximately 2-fold compared with the empty vector (EV) control to approximately 200 $\mu\text{g/L/OD}$ in the medium containing strains transformed with PUP-N and a Betv1 (Betv1 PUP-N) owing to the thebaine-forming activity of Betv1.

Example 15—General Methods

(277) The follow methods were used throughout the examples described above.

(278) Recombinant protein expression in *Escherichia coli*. All genes expressed in *E. coli* were codon-optimized, a His.sub.6-tag was added independently to the N-terminus and C-terminus, and the synthetic genes were cloned into the pACE vector. Expression vectors were transformed into *E. coli* stain Rosetta (DE3) pLysS (EMD Chemicals), which were subsequently induced overnight using 1 mM (w/v) isopropyl β -D-thiogalactoside (IPTG) at 16° C. Cells were harvested by centrifugation and sonicated in 50 mM sodium phosphate, pH 7.0, 300 mM NaCl and 10% (v/v) glycerol. After centrifugation at 20,000 g for 10 min, the supernatant was loaded onto Talon (Clontech) cobalt-affinity resin. Purification was performed according to the manufacturer's instructions. Purified, recombinant proteins were desalted using an Amicon (Millipore) centrifugal filter with a 30 kDa molecular weight cutoff, and stored in 50 mM sodium phosphate, pH 7.0, 50 mM NaCl and 10% (v/v) glycerol. Protein concentration was determined by the Bradford assay (Bio-Rad) using bovine serum albumin as the standard.

(279) Cross-linking experiments. Bis[sulfosuccinimidyl] suberate (BS3) was used as cross-linking

reagent and prepared in PBS buffer at a concentration of 10 mM, of which 2 μ L was added to 20 μ L of purified recombinant protein (1 μ g/ μ L). After incubation on ice for 1 h, 5 μ L of 5 $\times$ SDS (10% w/v) was added to quench the reaction. Samples were then boiled for 5 min and subjected to SDS-PAGE and immunoblot analysis.

(280) Enzyme assays. SalAT-coupled THS assays were performed in a 50-mL reaction containing latex protein or purified recombinant THS, 10 μ M salutaridinol, 50 μ M acetyl-CoA and 0.2 μ g SalAT in 10 mM sodium phosphate, pH 7.0, at 30° C. for 30 min. The assay was quenched with 200 μ L of acetonitrile and centrifuged at 17,000 g for 40 min. The supernatant was subjected to liquid chromatography-tandem mass spectrometry (LC-MS/MS) analysis. Direct THS assays were performed in a 50-mL reaction containing purified recombinant THS, phosphate buffer, pH 7.0, and salutaridinol-7-O-acetate at 30° C. for 30 sec. Chloroform (200 μ L) was then added to stop the reaction and protect any residual salutaridinol-7-O-acetate from hydrolysis.

(281) Liquid chromatography-tandem mass spectrometry (LC-MS/MS). Samples were analyzed using a 6410 Triple Quadrupole LC-MS (Agilent Technologies). Liquid chromatographic separation was performed using a Poroshell 120 SB-C18 HPLC column (Agilent Technologies) with a flow rate of 0.6 ml/min and a gradient of solvent A (0.05% [w/v] ammonium hydroxide, pH 6.5) and solvent B (100% acetonitrile) as follows: 0% solvent B from 1 min, 0-20% solvent B from 1 to 3 min, isocratic 99% solvent B from 6 to 8 min, 99-0% solvent B from 8 to 8.1 min, 0% solvent B from 8.1 to 11.1 min. Full scan mass analyses (m/z range 50-200) and collisional MS/MS experiments were performed as described previously (Nature Chemical Biology 11, 728-732, 2015).

(282) Virus-induced Gene silencing (VIGS). Two regions conserved in THS genes, but distinct from similar Betv1 homologs were selected as the VIGS target. The first fragment was located in the middle of the coding region and the second fragment included the 3' end of the open reading frame and part of the 3' untranslated region (FIG. 12A). The two fragments were synthesized as a one DNA fragment and cloned into the pTRV2 vector. The pTRV2 and the pTRV2-THS vectors were independently mobilized in *Agrobacterium tumefaciens*, which were subsequently cultured and infiltrated into opium poppy seedlings as described previously (Nature Chemical Biology 11, 728-732, 2015). Latex and stem samples were collected from ~30 mature plants. To confirm infiltration with *A. tumefaciens*, PCR was performed with a TRV2-MSC (multiple-cloning site) primer pair (FIG. 12B) to detect the occurrence of a mobilized fragment of the pTRV2 vector using cDNA isolated from individual stem samples. Alkaloids were extracted in acetonitrile from lyophilized latex and analyzed by LC-MS/MS. Relative transcript abundance of THS2 (SEQ ID NO. 6) in VIGS plant was determined by qRT-PCR using gene-specific primers (FIG. 12B).

(283) RNA extraction, cDNA synthesis and qRT-PCR. Total RNA was extracted using cetyl trimethyl ammonium bromide (CTAB) from frozen opium poppy tissue samples finely ground using a TissueLyser (Qiagen). cDNA synthesis was performed in a 10- μ L reaction containing approximately 1 μ g of total RNA using All-in-One RT mastermix (ABM) according to the manufacturer's instructions. SYBR-green qRT-PCR was used to quantify gene transcript levels. The 10- μ L reactions contained 1 $\times$ PowerUp SYBR Green master mix (Applied biosystems), 500 nM of each primer, and 2 μ L of a 20-fold diluted cDNA sample. A thermal profile of 50° C. for 2 min, 95° C. for 2 min, 40 cycles of 95° C. for 1 sec, and 60° C. for 30 sec (with a dissociation curve at the end) was used to perform qRT-PCR on a QuantiStudio Real-Time PCR System 3 (Applied Biosystems). Gene-specific primers were used for all qRT-PCR experiments (FIG. 12B). All primer pairs used in qRT-PCR were tested using amplicon dissociation curve analysis (95° C. for 15 sec at a ramp rate of 1.6° C./sec, 60° C. for 1 min at a ramp rate of 1.6° C./sec, and 95° C. for 15 sec at a ramp rate of 0.15° C./sec) to confirm the amplification stringency.

(284) Protein preparation for proteomics analysis. Soluble protein (~0.3 mg) was precipitated overnight using 3 volumes of ice-cold TCA/acetone (13.3:100, vol/vol) at -20° C. The protein pellet was collected by centrifugation at maximum speed for 20 min at 4° C. and washed three

times with ice-cold 75% (v/v) acetone. Protein pellets or gel bands excised from SDS-PAGE were submitted to Southern Alberta Mass Spectrometry centre at the University of Calgary for proteomics analysis.

Claims

1. A genetically modified cell comprising a recombinant polynucleotide encoding a heterologous thebaine synthase polypeptide wherein the thebaine synthase polypeptide comprises an amino acid sequence having at least 90% sequence identity to SEQ ID NO. 6, 31, 32, 80, or 81.
 2. The genetically modified cell of claim 1, wherein the thebaine synthase polypeptide converts salutaridinol-7-O-acetate to thebaine.
 3. The genetically modified cell of claim 1, wherein the thebaine synthase polypeptide comprises an amino acid sequence having at least 95% sequence identity to SEQ ID NO. 6, 31, 32, 80, or 81.
 4. The genetically modified cell of claim 1, wherein the thebaine synthase polypeptide comprises the amino acid sequence of SEQ ID NO. 6, 31, 32, 80, or 81.
 5. The genetically modified cell of claim 1, wherein the polynucleotide comprises the nucleotide sequence of SEQ ID NO. 19.
 6. The genetically modified cell of claim 1, wherein the cell is a plant cell, fungus, yeast, or bacterium.
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